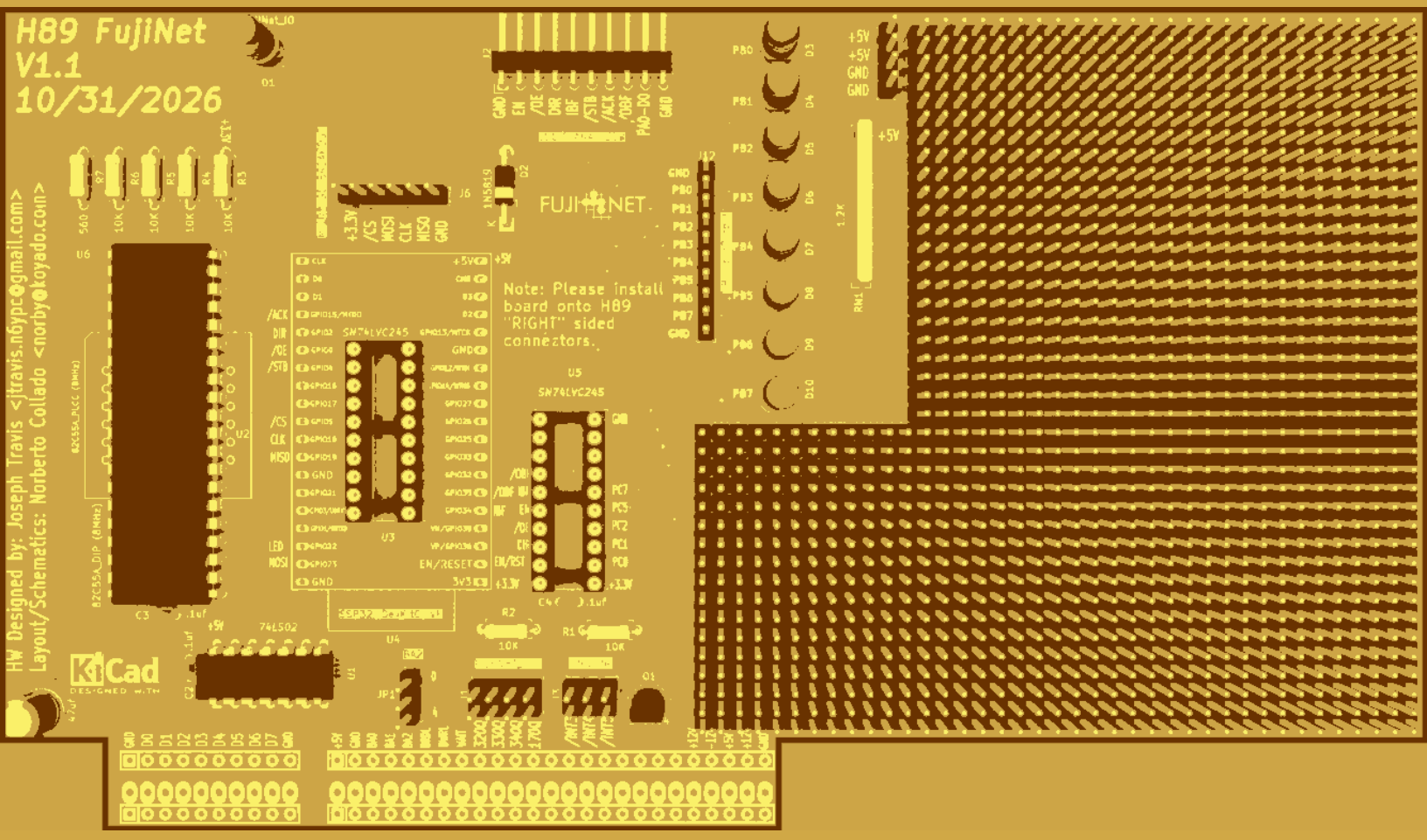
FUJINET MANUAL

for the

NETWORK ADAPTER

Model H89 FujiNet

OPERATION FN-89-01



FUJINET PROJECT DIRECTORY

The following addresses go straight to the people and things listed:

The FujiNet project, news and downloads	fujinet.online
FujiNet-Flasher (firmware updates)	fujinet.online/download
Help from other owners, any hour (Discord)	discord.gg/7MfFTvD
Source code for everything in this manual	github.com/FujiNetWIFI
The FNDISK drivers for HDOS and CP/M	github.com/FujiNetWIFI/fujinet-h89
The FujiNet Wiki	github.com/FujiNetWIFI/fujinet-firmware/wiki
Corrections to this manual	github.com/FujiNetWIFI/fujinet-manuals

YOUR FUJINET GUARANTEE

If you are not satisfied with your FujiNet – the card, its programs, or this manual – tell us. There is no Director of Customer Services, but there are a great many owners who will make certain your problem receives immediate, personal attention, usually within the hour, and usually from someone who has already met it on their own bench.

Our guarantee is a short one, and needs no attorney. Every part of the FujiNet is yours to see: the schematics, the firmware, CONFIG, the FNDISK drivers and every program printed in this manual are free software, published under the GNU General Public License, version 3. You may read them, build them, change them and give them away, as long as you pass the same freedom along.

The FujiNet is provided as it is, without warranty of any kind. We are not responsible for incidental or consequential damages, for a cold solder joint you made yourself, or for the hours you will lose browsing disk images you had forgotten you owned. We are, however, always glad to hear what you did with it.

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FujiNet Manual

for the

NETWORK ADAPTER

Model H89 FujiNet

OPERATION FN-89-01

FUJINET PROJECT
FUJINET.ONLINE

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Thomas Cherryhomes
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INTRODUCTION

Your FujiNet is a network adapter for the Heathkit H89 Computer. It is a single accessory circuit board that plugs into one of the accessory sockets on the CPU logic circuit board, beside the floppy disk interface, and it gives your Computer something no H89 had in 1979: a wireless connection to the world's computers, and to the disk images on them.

To HDOS and to CP/M, the FujiNet looks like more disk drives. You choose a disk image – from the microSD card on the board, from a file server on your own network, or from one of the public FujiNet servers on the Internet – and put it in one of eight FujiNet drives. Then you use it the way you use the diskette in drive SY0: or A:. You can catalog it, copy to it, run programs from it and keep your work on it.

Some features of the FujiNet are:

- Eight FujiNet drives. Under HDOS they are FN0: through FN7:. Under CP/M they are D: through K: (or wherever you choose to put them).
- Eight host slots, each naming a place disk images come from: the microSD card, a TNFS server, or a web, FTP, SMB or NFS server.
- Wireless networking on 2.4 GHz WiFi, with the network's name and password kept on the FujiNet, so you set them up only once.
- A microSD card socket on the board, for your own library of disk images.
- FujiNet CONFIG, a full-screen program for the H19 terminal, to join your network, browse hosts and put disk images in drives. There is a version for HDOS and one for CP/M.
- FNDISK, the drivers that let HDOS and CP/M use the FujiNet drives: FN.DVD for HDOS 2.0 and FNDISK.COM for Heath/Zenith CP/M 2.2.04.
- A web page of its own, on your network, for settings you would rather make from another computer.
- An acceptance test you key into the monitor with the Substitute command, so you can check the card with nothing more than the H: prompt.
- Room to grow. The FujiNet also carries a printer, a modem, eight network channels and a clock, for H89 programs to use.

The FujiNet does not replace your floppy disk drives. The H89 still starts HDOS or CP/M from a diskette in its own drive. Once your system is running, CONFIG and FNDISK bring in the FujiNet drives beside the ones you already have. See "FujiNet Software" on Page 5-1.

This Manual takes you through the whole job in order. You set the card's jumpers, install it, and check it with the acceptance test ("Set-Up and Testing," Page 3-1). You put the FujiNet programs on a diskette and prepare your system for them ("FujiNet Software," Page 5-1). Then you join your network and choose disk images with CONFIG (Page 6-1), and use the FujiNet drives from HDOS and CP/M (Page 7-1). The chapters after those cover the microSD card and the web page, firmware updates, troubleshooting, and how the card works.

ABOUT THIS MANUAL

STEPS

Every step you perform is marked with a pair of parentheses, like this:

( ) Do the thing the step tells you to do, then make a check mark between the parentheses.

If you are interrupted, the check marks show where you left off.

WHAT YOU TYPE

When this Manual shows you something to type, it is underlined, and set in the same characters the H19 shows you. The symbol ↵ means "push the RETURN key." For example:

H: G040100↵

means that, at the monitor's H: prompt, you type G, 0, 4, 0, 1, 0, 0 and then push RETURN. The monitor finishes the words it knows, so the screen will show "H: Go 040100." The parts the Computer types for itself are not underlined.

Keys are named by the words on their caps: **RETURN**, **ESC**, **TAB**, **ERASE**, **SHIFT**, **RESET**. "Push **SHIFT** and **RESET**" means hold down the right-hand SHIFT key, push RESET, and let go of both.

NUMBERS

The Heath monitor works in octal. As in all Heath manuals, an address is written in split octal, two octal bytes side by side: 040100 is the high byte 040 followed by the low byte 100 (2040 in hexadecimal). Port numbers are octal too. The FujiNet card answers at ports 320 through 323, which are D0 through D3 hexadecimal.

PICTORIALS AND PAGES

Drawings are called Pictorials and are numbered by chapter. Page numbers are by chapter as well: Page 6-3 is the third page of Chapter 6.

WHICH SOFTWARE

This Manual describes the first release of the FujiNet software for the H89. The FujiNet's programs are improved all the time. If something on your screen differs from a picture in this Manual, believe your screen, and look for news on the FujiNet Wiki (see the inside front cover).

CHANGES IN THIS EDITION

This second edition covers the V1.1 card, reworked for the present firmware (fujinet-firmware 7e8e647, October 2026). The first edition covered the V1.0 card. What changed:

- The card. The V1.1 board has a second 74LVC245 (U5) in place of the divider resistors, the 82C55A as U2 (a PLCC part) or U6 (a DIP), eight lamps on port B, and JP1 (BA2) beside PORT SEL. Every component number has changed. Pictorial 3-1, the jumper steps, the circuit description and the parts list are new.
- The rework. The ESP32's EN, IO0 and IO2 are no longer connected to the card. /OE and DIR reach it on IO21 and IO22, and LED D1 is gone (Chapter 11).
- Starting. The FujiNet runs from the moment the Computer is turned on. The Computer no longer starts it, and **SHIFT** and **RESET** no longer stop or restart it: the FujiNet drives stay in service through a reset (Chapters 4 and 6).
- Firmware updates are done over USB alone (Chapter 9). The monitor commands and the program that put the FujiNet in its loading mode are gone.
- Testing. The Port Test writes a pattern to port B instead of starting the FujiNet, and the Communication Test is 114 numbers long instead of 129 (Chapter 3).

SPECIFICATIONS

THE CARD

Bus	Heath H89/H88 accessory bus. Uses one pair of accessory sockets on the CPU logic circuit board: P504 with P510, or P505 with P511.
I/O ports	320 through 323 octal (D0–D3 hexadecimal), selected with the PORT SEL jumper (J1) and JP1.
Other PORT SEL positions	330, 340 and 170 octal. Not used by the present software.
Interface	82C55A programmable peripheral interface. Port A in Mode 2 (bidirectional, strobed) carries the data; port C carries control and status.
Interrupts	None. J3 is left open.
Level shifting	Two 74LVC245 bus transceivers: one for the data, one for the control lines.
Reset	The H89's RESET resets the 82C55A only. The FujiNet runs from power-on and keeps running through a reset.
Lamps	Eight LEDs (D3–D10) on the 82C55A's port B, for programs to use.

THE FUJINET

Processor	Espressif ESP32, two cores at 240 MHz, on an ESP32-DevKitC-VE module.
Memory	8 megabytes of flash (firmware and settings) and 8 megabytes of PSRAM.
Wireless	IEEE 802.11 b/g/n, 2.4 GHz band only. Open, WPA and WPA2 personal networks.
Storage	microSD card socket. Cards formatted FAT32; 32 gigabytes or smaller recommended.
USB	On the ESP32 module. The only way to update the firmware; also the diagnostic output.

DEVICES THE FUJINET PROVIDES

Disk drives	8 drive slots.
Hosts	8 host slots: the microSD card, TNFS servers, and web (HTTP, HTTPS), FTP, SMB and NFS servers.
Network channels	8 (N1 through N8): TCP, UDP, HTTP and HTTPS, TNFS, FTP, SMB, NFS, SSH, TELNET and more.
Printer	1. Pages are collected on the FujiNet's web page.
Modem	1, Hayes compatible.
Clock	Network time.

FUJINET DRIVES

Disk image format	H8D: 102,400 bytes, 40 tracks of 10 sectors of 256 bytes. This is the H-17-1 single-sided diskette, sector for sector.
Under HDOS	FN.DVD device driver. Units FN0: through FN7:. Catalog, read, write, mount and dismount.
Under CP/M	FNDISK.COM. Drives D: through K:, or eight drives starting at a letter you choose. 92K and 64 directory entries per drive.
Starting the system	From the H89's own floppy drive. The FujiNet drives are added once HDOS or CP/M is running.

SOFTWARE

Operating systems	HDOS 2.0. Heath/Zenith CP/M 2.2.04 (a 62K system).
FujiNet programs	CONFIG (CONFIG.ABS for HDOS, CONFIG.COM for CP/M), FN.DVD and FNDISK.COM.
Monitors	The acceptance test runs on MTR-88, MTR-89 and MTR-90.

MEMORY REQUIRED

HDOS	48K; 64K to run CONFIG after the FujiNet drives have been used.
CP/M	64K, to run a 62K CP/M system.

POWER AND SIZE

Power	From the H89's bus: +5 volts, and through D2 to the ESP32 module, whose regulator makes 3.3 volts. No separate power supply.
Current	(to be measured)

Dimensions..... (to be measured)

SET-UP AND TESTING

WHAT YOU NEED

Before you begin, make sure you have:

- Your Heathkit H89 (or H88, or Zenith Z-89), working, with its monitor ROM: MTR-88, MTR-89 or MTR-90. The Computer needs at least 48K of memory for HDOS (64K is better; see Page 4-1), and 64K for CP/M.
- A system diskette that starts HDOS 2.0 or Heath/Zenith CP/M 2.2.04, and a blank diskette for the FujiNet programs (see "FujiNet Software," Page 5-1).
- The name and password of your WiFi network. The FujiNet uses the 2.4 GHz band only.
- A microSD card, if you want a library of disk images on the board itself. It is not needed to begin with.
- A small flat-blade screwdriver, for the cabinet latches.

PARTS LIST

Check the contents of your package against this list.

QTY.	DESCRIPTION
1	H89 FujiNet circuit board (V1.1, reworked; see "The Rework" in Chapter 11), assembled and tested, with the FujiNet firmware loaded
2	Jumpers (shunts), installed on the PORT SEL header at 320Q and on JP1 at 0

If anything is missing, see "Getting Help" in the Appendix.

HANDLING THE CARD

CAUTION: The integrated circuits on the FujiNet card are MOS devices, and can be damaged by static electricity. Before you pick up the card, touch the metal chassis of the Computer (with the line cord unplugged) to equalize the static charge between you and it. Then hold the card by its edges. Do not touch the pins of the edge connectors, the pins of the ESP32 module, or the module's antenna.

Keep the card in its protective bag until you install it. If you must set it down, lay it on the bag, component side up.

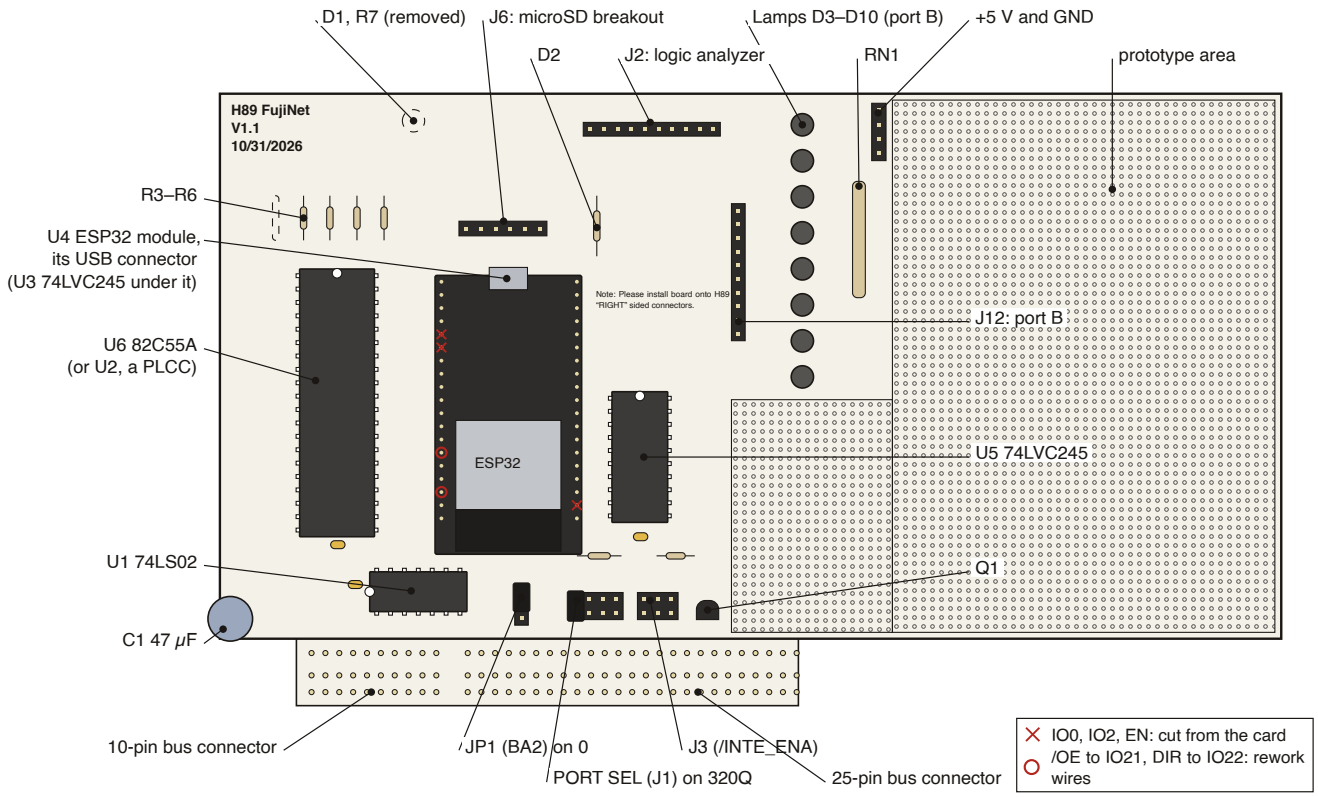
SETTING THE JUMPERS

Refer to Pictorial 3-1 for the location of the jumpers, and to Pictorial 3-2 for the following steps.

The PORT SEL jumper chooses the group of eight I/O ports the card answers, and JP1 chooses the first four ports of the group or the last four. The FujiNet programs expect the card at port 320, so the PORT SEL jumper goes on the 320Q position and JP1 on 0. The other positions are for future software.

J3 connects the card to one of the Computer's interrupt lines. The FujiNet programs do not use interrupts, so J3 has no jumper on it.

( ) Find the PORT SEL header, J1 (marked IO_PORT_SEL), near the bottom edge of the card, below the ESP32 module. Make sure the jumper is on the pair of pins marked 320Q, and on no other.



PICTORIAL 3-1
The H89 FujiNet card, component side.

( ) Find JP1, marked BA2, to the left of J1. Make sure its jumper joins the middle pin to the pin marked 0.

( ) Find J3, marked /INTE_ENA, to the right of J1. Make sure there are no jumpers on it. If there is one, remove it and store it on a single pin of J3 so it is not lost.

JUMPER	TO
PORT SEL (J1)	320Q
JP1 (BA2)	0
J3 (/INTE_ENA)	No jumper

NOTE: Port 320 is also the second port of the HA-88-3 three-port serial interface accessory, which HDOS uses as its alternate terminal port. If your Computer has an HA-88-3 board with an 8250 IC installed for that port, it and the FujiNet would both answer at 320. See “Installing the Card.”

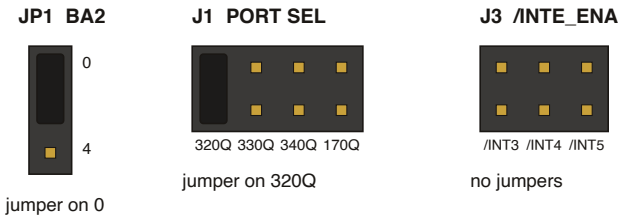
THE MICROSD CARD

You can put a microSD card in the FujiNet’s socket to keep your own library of disk images on the board. The card must be formatted FAT32. Cards of 32 gigabytes or smaller are formatted that way when new.

( ) On your personal computer, copy your .H8D disk images onto the microSD card. You may put them in folders.

( ) Slide the microSD card into the socket on the breakout board near the top edge of the FujiNet card, contacts first and label up, until it stops.

The FujiNet looks at the microSD card only when it starts, when the Computer is turned on. Insert or remove the card only while the Computer is turned off, and with no USB cable plugged into the ESP32 module: a USB cable keeps the FujiNet running.



PICTORIAL 3-2
PORT SEL at 320Q; no jumper on J3.

CABINET REMOVAL

Whenever you need to remove the cabinet shell:

- Turn off the Computer and unplug the line cord.
- Insert the blade of a small screwdriver into the notch in the latch plate on one side of the cabinet, and slide the latch plate toward the back of the Computer about 1/2″.
- Likewise, open the latch plate on the other side of the cabinet shell.
- Carefully tilt the cabinet shell back. Never allow the cabinet shell to hang unsupported.

WARNING: When the line cord is connected to an AC outlet, hazardous voltages can be present inside your Computer, even with the power switch off. Always unplug the line cord before you open the cabinet.

Simply reverse this procedure to close and lock the cabinet shell.

PHOTOGRAPH TO COME
The H89 with its cabinet shell tilted back, showing the CPU logic circuit board.

PICTORIAL 3-3
The cabinet shell tilted back.

INSTALLING THE CARD

Refer to Pictorial 3-4 for the following steps.

The CPU logic circuit board has three pairs of accessory sockets. Each pair is a 10-pin socket and a 25-pin socket: P504 with P510, P505 with P511, and P506 with P512. The floppy disk interface board uses P506 and P512, and must stay there. The FujiNet card goes in either of the other pairs.

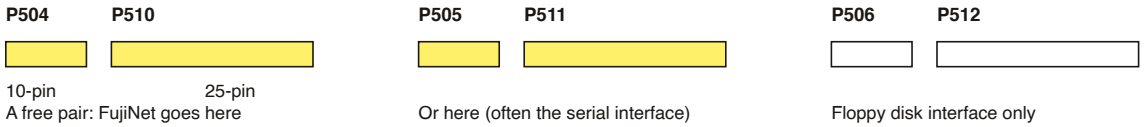
( ) Open the cabinet. See “Cabinet Removal.”

( ) Find the three pairs of accessory sockets on the CPU logic circuit board, and note which boards are already installed.

( ) Choose a free pair: P504 with P510, or P505 with P511.

( ) If an HA-88-3 serial interface board is installed, look at it. If the 8250 IC for its 320 port is in its socket, the serial port and the FujiNet cannot share the Computer. Remove that one 8250 (handle it as a protected IC), or do not install the FujiNet card.

( ) Hold the FujiNet card by its edges, component side facing the same way as the other accessory boards. Line up its 10-pin connector with the 10-pin socket (P504 or P505) and its 25-pin connector with the 25-pin socket (P510 or P511).



PICTORIAL 3-4
The accessory sockets on the CPU logic circuit board (not to scale).

( ) Press the card straight down onto the sockets, evenly, until it is fully seated. Make sure every pin went into its socket, and that no row is offset by one.

( ) Make sure no wires or cables lie across the ESP32 module's antenna.

( ) Close and lock the cabinet shell, and plug in the line cord.

NOTE: The FujiNet's firmware is updated through the USB connector on the ESP32 module. If you expect to update it, you may want to plug a USB cable into the module now and lead it out of the cabinet, so you need not open the Computer later. See "Firmware Updates" on Page 9-1.

The FujiNet's antenna is inside the cabinet. The plastic shell lets the signal through, but the picture tube and chassis do not. If your WiFi router is far away, you may need to move it closer.

PHOTOGRAPH TO COME
The FujiNet card seated on P504 and P510, beside the floppy disk interface board.

PICTORIAL 3-5
The FujiNet card installed.

TESTING

The purpose of the "Testing" is to verify that your FujiNet card is installed and working properly. It is not necessary, at this time, to have a working knowledge of the FujiNet, nor to have set up your WiFi network: the tests use nothing but the Computer's monitor. If, at any time during the "Testing," you fail to obtain the proper results, refer to the "Troubleshooting" section on Page 10-1.

There are three tests. The Initial Tests make sure the Computer still starts with the card in it. The Port Test (for MTR-90 only) checks the card's 82C55A with the monitor's own commands. The Communication Test is a short program you key into memory; it sets up the card, sends the FujiNet a request, and prints the answer.

INITIAL TESTS

( ) Push the rear panel POWER switch to ON. The speaker should produce a "beep," and after the picture tube warms up the monitor's prompt, H:, appears at the left of the screen.

NOTE: If your Computer starts HDOS or CP/M by itself at power-up, or runs a memory test first, switch SW501 on the CPU logic circuit board is set for it. Push **SHIFT** and **RESET** to stop it and get the H: prompt, or see Page 10-1.

( ) Push the right-hand **SHIFT** key and the **RESET** key together. The Computer should beep and give you the H: prompt again.

If the prompt does not appear, turn the Computer off at once and see "Troubleshooting."

PORT TEST (MTR-90)

The MTR-90 monitor has an Out command, which writes a number to an I/O port, and an In command, which reads one. If your monitor shows "Out" when you type O, you can use them to check the card's 82C55A with no program at all. If your monitor is MTR-88 or MTR-89, skip to the Communication Test.

( ) Type O the monitor finishes the word "Out." Type 323,300↵. This sets up the 82C55A.

( ) Type I the monitor finishes the word "In." Type 322↵. The monitor shows what it read from port 322 on the next line. It should be 200.

( ) Type O again, and then 321,252↵. This writes a pattern to port B, which lights every other one of the card's eight lamps.

( ) Type I321↵. The monitor should show 252, the pattern you wrote.

```
  H: Out 323,300
  H: In 322
200
  H: Out 321,252
  H: In 321
252
```

The first number shows the 82C55A answered at port 322. (The second digit may be 4 instead of 0: 240 is a good result too.) The second shows the 82C55A keeps what is written to it. If you see 377, nothing answered: check the PORT SEL jumper, JP1, and that the card is seated.

( ) Push **SHIFT** and **RESET**. This resets the 82C55A. The FujiNet itself keeps running.

COMMUNICATION TEST

This test is the Computer and the FujiNet talking to each other. You will first enter a short program into memory with the monitor's Substitute command, and then run it. The program sets up the card's 82C55A, asks the FujiNet for its adapter configuration, and prints what comes back.

The program prints a 1 as soon as it finds the card's 82C55A, and a 2 once the FujiNet has taken the request. Then it prints the text in the FujiNet's answer: the name of your WiFi network and the FujiNet's own name, if it has joined a network, and the version of its firmware. Then the monitor beeps, prints an H, and gives you the prompt.

The program is 114 numbers long. Take your time; if you make a mistake, you can go back and fix it.

( ) Push **SHIFT** and **RESET** to make sure you have the H: prompt.

( ) Type S the Computer will finish the word "Substitute." Type 040100↵. The Computer will respond with:

```
040100 XXX
```

Each X can be any digit, 0 through 7. It is what is in memory now.

( ) Enter 076 and push the SPACE bar. The Computer moves to the next address:

```
040101 XXX
```

( ) Enter 300 and push the SPACE bar. Then continue by making the following entries, eight to a line in the table. Push the SPACE bar once after each entry. The address at the left of each line of the table is the address the Computer should be showing when you begin that line; check it as you go.

```
ADDRESS  ENTER
040100    076 300 323 323 333 322 346 007
040110    302 251 040 076 001 323 322 076
040120    061 315 302 003 041 252 040 006
040130    010 176 323 320 333 322 346 200
040140    312 134 040 043 005 302 131 040
040150    076 000 323 322 076 062 315 302
040160    003 001 000 005 315 237 040 376
040170    300 312 161 040 005 302 164 040
040200    315 237 040 376 300 312 251 040
040210    376 041 332 222 040 376 177 332
040220    230 040 076 040 271 312 200 040
040230    117 315 302 003 303 200 040 333
040240    322 346 040 312 237 040 333 320
040250    311 166 300 160 350 006 000 137
040260    000 300

  H: Substitute 040100
040100 000 076
040101 000 300
040102 000 323
040103 000 323
040104 000 333
040105 000 322
```

( ) After the last entry (300 at address 040261), push RETURN instead of the SPACE bar. The Computer gives you the H: prompt.

If you made a mistake: while you are still entering, push the minus sign (-) to go back one address, and type the number again. Only the last three digits you type count. To check your work afterward, type S040100↵ and push the SPACE bar to step through the numbers without changing them; push RETURN when you are done.

( ) Type G the Computer will finish the word "Go." Type 040100↵.

At once, the display should be:

```
12 Workshop FujiNet v1.6-6b818770 H
  H:
```

The text after 12 is the FujiNet's answer. Here, "Workshop" is the WiFi network the FujiNet has joined, "FujiNet" is its own name, and v1.6-6b818770 is its firmware version. Until the FujiNet has joined a network, the names are blank and only the firmware version shows. You may also see a few stray letters and marks between the names: they are the bytes of the FujiNet's network addresses that happen to be printable characters.

The program stays in memory until you turn off the Computer. To run it again, push **SHIFT** and **RESET** and type G040100↵. (Typing G with no address would not start it over.)

WHAT THE TEST TELLS YOU

YOU SEE	IT MEANS
12 and text, H	The card is working. Go on to "System Configuration."
Only H (and a beep)	No 82C55A answered at port 320: the PORT SEL jumper, the card's seating, or another board at 320.
1, and nothing more	The 82C55A is there, but the FujiNet is not taking the request: its firmware, its power, or the 74LVC245s.
12, and nothing more	The request went out but no answer came back: the FujiNet's firmware, or the return path through the 82C55A.
No beep, no H, or a garbled screen	The program was keyed in wrong. Push **SHIFT** and **RESET** and check your entries.

For more about each of these, see "Troubleshooting" on Page 10-1. The program itself is listed in the Appendix.

NOTES:

1. If you have not yet made your FujiNet diskette, proceed to "FujiNet Software" (Page 5-1).
2. If your FujiNet diskette is ready, proceed to "System Configuration" (Page 4-1) and then "FujiNet CONFIG" (Page 6-1).

SYSTEM CONFIGURATION

THE FUJINET IN YOUR SYSTEM

This chapter of the Manual shows how the FujiNet card fits into your Computer: the I/O ports it uses, the memory its programs need, and the settings of its jumpers. If you have set the jumpers as described in "Set-Up and Testing" and the Communication Test passed, nothing here needs changing. Read it before you add other accessories, or if you are using your Computer with non-Heath equipment.

I/O PORTS
The card answers at four I/O ports, 320 through 323 octal (D0 through D3 hexadecimal). The 82C55A's port A, at 320, carries the bytes between the Computer and the FujiNet in both directions; its port C, at 322, carries the handshake and the lines that steer the bus transceiver; 323 is the 82C55A's control register. Port 321 (port B) drives the card's eight lamps, D3 through D10, and the header J12; the FujiNet does not use it.

The Computer's address decoder gives each accessory a group of eight ports, and the PORT SEL jumper picks the group. JP1 picks the first four ports of the group (on 0) or the last four (on 4).

The card uses only four ports of its group, but no other accessory may use the same group. In a standard Heath system, the group 320–327 belongs to the second port of the HA-88-3 serial interface (the DCE port HDOS uses as its alternate terminal). It is free unless that port's 8250 is installed.

MEMORY
The FujiNet uses no memory in the Computer. Its programs do:

- HDOS: CONFIG.ABS runs from 052200 (2A80 hexadecimal) up, and FN.DVD, the HDOS device driver, takes about 4.3K at the top of memory once it is loaded. A 64K Computer (HDOS says "56K") has room for both. A 48K Computer has room for CONFIG only until FN.DVD is loaded; after that, HDOS answers "NOT ENOUGH RAM FOR THIS PROGRAM" until you start the system again. On a 48K Computer, run CONFIG first, then use the FujiNet drives.
- CP/M: FNDISK.COM puts about 2K of driver above CP/M's BIOS, in the space a 62K system leaves free in a 64K Computer. See "Making a 62K CP/M System" on Page 5-3.

INTERRUPTS
The FujiNet programs do not use interrupts; they watch the 82C55A's status instead. Leave J3 with no jumper. A jumper on J3 would tie the card to one of the Computer's interrupt levels, 3, 4 or 5, which the console and other accessories already use.

RESET
The Computer's RESET line resets the card's 82C55A, and nothing else on the card. So whenever you push **SHIFT** and **RESET**, the Computer loses touch with the FujiNet until the next FujiNet program you run sets the 82C55A up again, which it does at once. The FujiNet itself keeps running, with its drives in service.

The FujiNet starts afresh only when the Computer is turned on. Its settings, its host slots and your choices of disk images are always kept; but after it starts, the drives are not put back in service until you run CONFIG, unless you have told the FujiNet to do it by itself (see "When the FujiNet Starts" on Page 6-12).

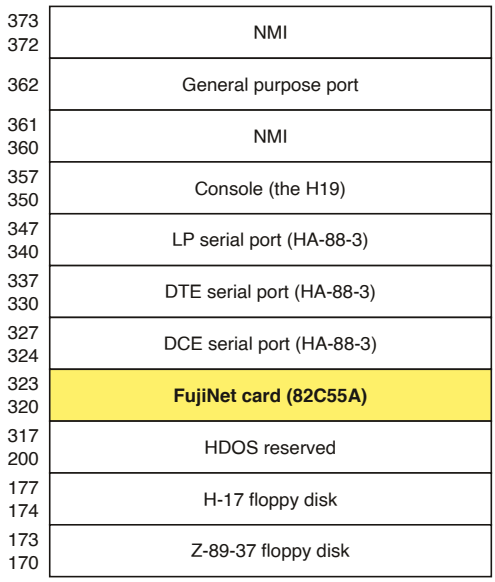
I/O PORT MAP

The I/O Port Map shows the ports in a typical Heath H89, octal, with the FujiNet card's four marked. The ports between 200 and 317 are reserved for HDOS. A Z-89-37 soft-sectored floppy disk interface uses 170–173; that is why the card's 170Q PORT SEL position is not used.

JUMPER SETTINGS

JUMPER	POSITION	USE
PORT SEL (J1)	320Q	The group 320–327. **The position to use.**
	330Q	The group 330–337 (the HA-88-3 DTE port's). Not used by present software.
	340Q	The group 340–347 (the HA-88-3 LP port's). Not used by present software.
	170Q	The group 170–177 (the Z-89-37's). Not used by present software.
JP1 (BA2)	0	The first four ports of the group: 320–323. **The position to use.**
	4	The last four: 324–327. Not used by present software.
J3	none	No interrupts. **The setting to use.**
	/INT3, /INT4, /INT5	Interrupt level 3, 4 or 5. Not used by present software.

NOTE: The FujiNet programs look for the card at 320 only. A card set to another position will not be found.



PICTORIAL 4-1
The I/O ports in an H89.

FUJINET SOFTWARE

THE FUJINET DISKETTES

The FujiNet's programs for the H89 come as two disk images, one for each operating system:

DISK IMAGE	HOLDS
FujiNet for HDOS	CONFIG.ABS and FN.DVD
FujiNet for CP/M	CONFIG.COM and FNDISK.COM

You can get both from the FujiNet project (see the inside front cover). Each is an .H8D image of a single-sided H-17-1 diskette, so it can be written onto a real diskette for your Computer.

Write the image for your operating system onto a blank hard-sectored diskette, with whatever means you already use to bring disk images to your H89: a program that sends them over a serial cable from a personal computer, a floppy drive emulator that accepts .H8D images, or a friend's machine. You need do this only once. When the FujiNet is running, you can copy any other disk image from a FujiNet drive onto a diskette with HDOS or CP/M itself.

Neither diskette starts the Computer by itself. You start HDOS or CP/M from your own system diskette as usual, and then run the FujiNet programs from the FujiNet diskette in your second drive. (You can also copy them onto your system diskette.)

THE PROGRAMS

PROGRAM	FOR	WHAT IT DOES
CONFIG.ABS	HDOS	FujiNet CONFIG: joins your WiFi network, sets up host slots, and puts disk images in drives. See Page 6-1.
FN.DVD	HDOS	The HDOS device driver for the FujiNet drives, FN0: through FN7:. See Page 7-2.
CONFIG.COM	CP/M	FujiNet CONFIG for CP/M.
FNDISK.COM	CP/M	Adds the FujiNet drives to CP/M as D: through K:. See Page 7-6.

Each operating system needs a little preparation before its FujiNet drives can be used. HDOS needs FN.DVD on its system diskette. CP/M needs to be made 2K smaller, so that FNDISK has room. Both jobs are described below and need to be done only once.

PREPARING HDOS

HDOS takes in the device drivers it finds on the system diskette (SY0:) as it starts. To make it take in the FujiNet drives, you copy FN.DVD onto your system diskette and start HDOS again.

CAUTION: Always work on a copy of your HDOS system diskette, never on the distribution diskette.

( ) Make a working copy of your HDOS 2.0 system diskette, and start HDOS from it.

NOTE: HDOS 2.0 cannot take a date after 1999. When it asks for the date, give it a year of 99 or earlier, in the form DD-MMM-YY.

( ) Put the FujiNet for HDOS diskette in drive 1, and type `MOUNT SY1:`↵.

( ) Type `COPY SY0:FN.DVD=SY1:FN.DVD`↵.

If HDOS answers "1 FILES COPIED," go on to the next step. If it answers "NO FREE SPACE ON MEDIA," as it will on a full distribution copy, make room. The stock system diskette carries DK.DVD, the driver for the H47 eight-inch disk system; if you have no H47, it is not needed.

( ) If there was no room, type `DELETE SY0:DK.DVD`↵ and then `COPY SY0:FN.DVD=SY1:FN.DVD`↵ again.

```
>MOUNT SY1:
VOLUME 000, MOUNTED ON SY1:
LABEL: FUJINET FOR HDOS
>COPY SY0:FN.DVD=SY1:FN.DVD
ERROR ON FILE SY0:FN.DVD - ?02 NO FREE SPACE ON MEDIA
>DELETE SY0:DK.DVD
>COPY SY0:FN.DVD=SY1:FN.DVD
1 FILES COPIED
```

( ) Push **SHIFT** and **RESET**, and start HDOS again. (When it offers the date it has, just push RETURN.)

( ) Type `STAT`↵. Under DEVICE STATUS, HDOS should list FN: with the other devices.

```
DEVICE STATUS
  TT:   IP      RW
  SY:   IP      DRW
  FN:           DRW
)
```

HDOS now knows the FujiNet drives. It loads FN.DVD the first time you use one of them, and keeps it until you start the system again.

You may also copy CONFIG.ABS onto your system diskette, if there is room, so that you can type RUN CONFIG ↵ without the FujiNet diskette:

```
)COPY SY0:CONFIG.ABS=SY1:CONFIG.ABS↵
```

MAKING A 62K CP/M SYSTEM

FNDISK puts a small driver, about 2K, in memory above CP/M's own BIOS. In a 64K CP/M system there is no memory free above the BIOS, so you make your CP/M 2K smaller, a "62K system," with the programs Heath supplies for the purpose. Your programs will have 2K less room, which few of them will notice.

FNDISK needs Heath/Zenith CP/M version 2.2.04, the one with MOVCPM17 for H-17 disks.

CAUTION: Always work on a copy of your CP/M system diskette, never on the distribution diskette.

( ) Make a working copy of your CP/M 2.2.04 system diskette, and start CP/M from it.

( ) Type MOVCPM17 62 *↵. MOVCPM17 builds a 62K CP/M in memory; it takes a few seconds, and then says it is ready for SYSGEN.

( ) Type SYSGEN↵. SYSGEN asks for a source drive. The new system is already in memory, so just push RETURN.

( ) SYSGEN asks for a destination drive. Type A. When it says "DESTINATION ON A, THEN TYPE RETURN," make sure the working copy is in drive A and push RETURN. SYSGEN writes the new system onto the diskette and says FUNCTION COMPLETE.

```
A>MOVCPM17 62 *
MOVCPM17 VERSION 2.2.04
CONSTRUCTING 62k CP/M vers 2.2
READY FOR "SYSGEN" OR
"SAVE 38 CPM62.COM"
A>SYSGEN
SYSGEN VER 2.0.04
SOURCE DRIVE NAME (OR RETURN TO SKIP):
DESTINATION DRIVE NAME (OR RETURN TO REBOOT):A
DESTINATION ON A, THEN TYPE RETURN
FUNCTION COMPLETE
DESTINATION DRIVE NAME (OR RETURN TO REBOOT):
```

( ) SYSGEN asks for another destination. Push RETURN, and CP/M starts again from the diskette.

The sign-on message now shows the new size:

```
62K HEATH/ZENITH CP/M 2.2.04  09/15/82
FOR H17 DISKS WITH OPTION(S) I
A)
```

This working copy is the system diskette to use with the FujiNet. Make another copy of it to keep.

FUJINET CONFIG

STARTING CONFIG

FujiNet CONFIG is the program you use to set up the FujiNet: to join your WiFi network, to tell the FujiNet where your disk images are, and to put disk images in its eight drives. It fills the H19's screen, and you work it from the keyboard. There is a version for each operating system; they look and work alike.

UNDER HDOS

( ) Start HDOS from your system diskette (the one with FN.DVD on it).

( ) If any FujiNet drives are mounted, dismount them first: `DISMOUNT FN0:`↵, and so on. (See Page 7-3.)

( ) Put the FujiNet for HDOS diskette in drive 1 and type `MOUNT SY1:`↵. Then type `RUN SY1:CONFIG.ABS`↵.

If you copied CONFIG.ABS onto your system diskette, `RUN CONFIG`↵ is enough.

NOTE: On a 48K Computer, CONFIG has room to run only before FN.DVD is loaded (by the first `MOUNT FN` or `LOAD FN:`). Run CONFIG first after you start HDOS; to run it again later, start HDOS again. A 64K Computer can run it at any time.

UNDER CP/M

( ) Start CP/M from your 62K system diskette.

( ) Put the FujiNet for CP/M diskette in drive B and type `B:CONFIG`↵.

WHAT YOU SEE FIRST

CONFIG draws its frame and says `Contacting FujiNet...` while it sets up the card and asks the FujiNet how things stand. Just after the Computer is turned on, that can take a few seconds, while the FujiNet starts. Then:

- If the FujiNet has never been told about a WiFi network, CONFIG shows the WiFi Networks screen. See "Joining Your WiFi Network."
- If it knows a network, CONFIG joins it, and shows the Host Slots and Drive Slots: the main screen, where you will spend most of your time.



PICTORIAL 6-1
CONFIG's main screen, under HDOS.

THE SCREEN

Every CONFIG screen has the same parts, as in Pictorial 6-1:

- The top line names the program and the operating system it is running under: H89 HDOS or H89 CP/M.
- The upper and lower panes hold lists: here, the eight host slots and the eight drive slots. The name of the pane you are working in is shown in reverse video, and the selected line of it carries a bright bar with an arrow at its left.
- The line under the panes is the information line. CONFIG asks its questions here, and tells you what it is doing.
- The two lines below the frame show the keys you can use now. A key's name is shown in reverse video: **RET** means the RETURN key.
- The H19's 25th line, at the very bottom, shows the WiFi network the FujiNet has joined, its address on that network, and the version of its firmware.

THE KEYS

CONFIG uses the H19's numeric keypad for moving about. It sets the keypad to its shifted mode while it runs, so the arrow keys move the bar even without SHIFT; it puts the keypad back when you leave.

KEY	DOES
8 / **2** (arrows)	Move the bar up and down. Down from host slot 8 goes into the drive slots.
4 / **6** (arrows)	In a list of files, turn to the previous or next page.
5 (HOME), **1** (IL)	Go to the first or last line.
9 (DC), **3** (DL)	Previous or next page of files.
1 to **8**	On the main keyboard: pick that slot.
RETURN	Choose the selected line.
TAB	Move between the host slots and the drive slots.
ESC	Go back a step. On the main screen: mount the drives and leave CONFIG.
ERASE	In a list of drives: eject them all. When typing: clear the line.
BACK SPACE	When typing: rub out the last character. In a file list: go up a folder.
Letters	As shown on the key lines: E, R, W, C, F, N, S, Q.

JOINING YOUR WIFI NETWORK

The FujiNet keeps the name and password of one WiFi network and joins it whenever it starts. You set them up once, with this screen. If CONFIG did not show it to you on its own, get to it from the main screen by pushing C (Info & WiFi) and then S (Change WiFi network).

CONFIG has the FujiNet look for networks, and lists those it can hear, strongest first, with a signal meter and the strength in dBm (nearer zero is stronger). The FujiNet hears only networks on the 2.4 GHz band.

( ) Move the bar to your network and push RETURN.

If your network does not announce its name, it will not be listed. Move the bar to ⟨Enter a network name⟩ at the bottom of the list, push RETURN, type the name exactly (capitals count) and push RETURN.

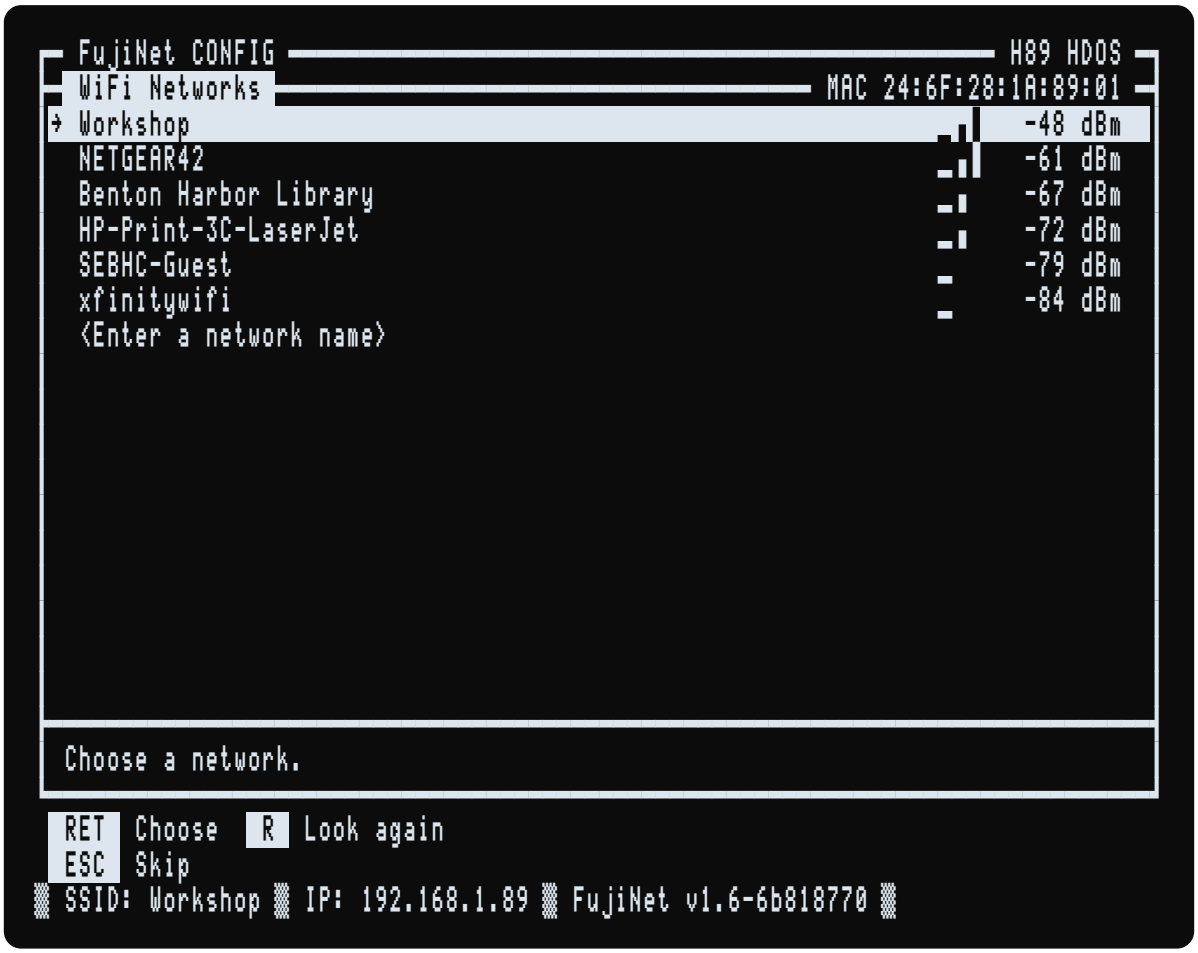
If you don't see your network, push R to look again. To leave this screen without joining a network, push **ESC**.

( ) CONFIG asks for the network's password. Type it, exactly, and push RETURN. CONFIG shows an asterisk for each character. If your network has no password, just push RETURN.



CONFIG then shows "Connecting to" and the network's name while the FujiNet joins it. The information line keeps you posted: PLEASE WAIT...(ESC TO ABORT), and then CONNECTION SUCCESS! and the main screen. If the FujiNet cannot join the network you will see CONNECT FAILED or UNABLE TO CONNECT, and CONFIG goes back to the list: check the password and try again. Joining can take up to 50 seconds; **ESC** gives up.

The FujiNet remembers the network. You will not see this screen again unless you ask for it, or the network goes away.



PICTORIAL 6-2
Choosing a WiFi network.

HOST SLOTS

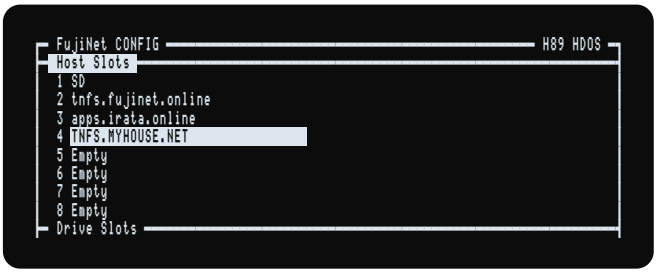
A host is a place disk images come from. The FujiNet has eight host slots, each holding the name of one host. Slot 1 is normally the microSD card, called SD, and two public FujiNet servers on the Internet are set up in slots 2 and 3:

```
1 SD
2 tnfs.fujinet.online
3 apps.irata.online
```

CHANGING A HOST SLOT

( ) On the main screen, pick the host slot with the arrow keys, or with its number on the main keyboard.

( ) Push E. The information line asks for the host name, and the slot's line turns to reverse video for you to type in. Type the host's name and push RETURN.



Push **ERASE** to clear the line and start over, or **ESC** to leave the slot as it was. Pushing RETURN on an empty slot starts the editing too.

What you type depends on the kind of host:

YOU TYPE	FOR
SD	The microSD card in the FujiNet.
tnfs.fujinet.online	A TNFS server (a plain name means TNFS).
http://host/path/	A web server's folder (https:// works too).
ftp://host/path/	An FTP server.
smb://host/share/	A Windows or Samba share.
nfs://host/export/	An NFS server.

A name can be up to 31 characters. If you change the name in a slot, any drives using disk images from that host are emptied.

THE PUBLIC SERVERS

tnfs.fujinet.online and apps.irata.online are run by FujiNet owners for everyone's use, and hold software for many computers in folders named for them; look in the H89 folder. Like all TNFS hosts on the Internet, they are read only: you can put their disk images in drives to read, but to write on one, copy it to your microSD card first (see "Copying a File," Page 6-9).

BROWSING A HOST

( ) On the main screen, pick a host slot and push RETURN.

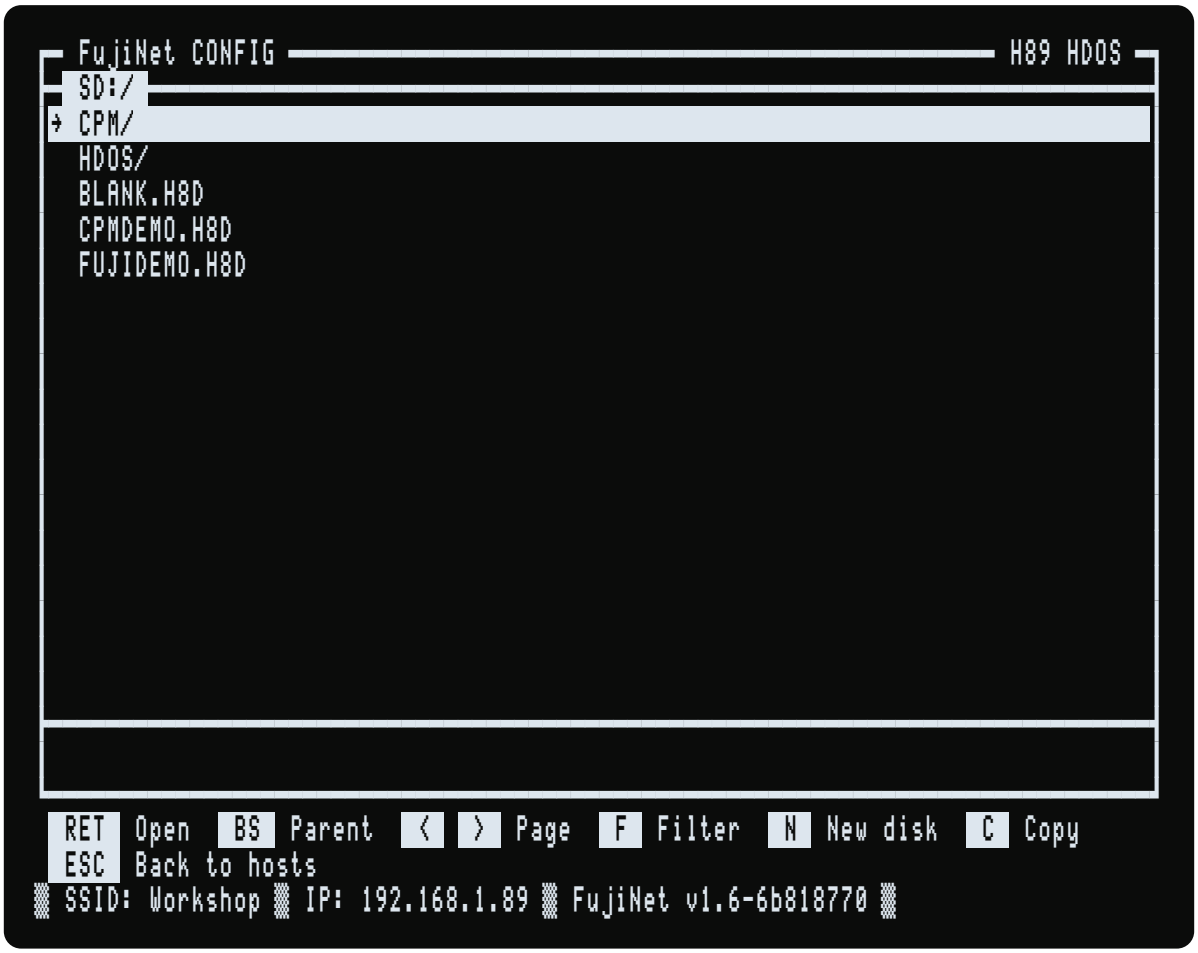
CONFIG has the FujiNet open the host and lists what is in it, a page of seventeen names at a time. Names ending in a slash (/) are folders. The top line of the pane shows the host and the folder you are in.

KEY	DOES
RETURN	Open the folder, or choose the disk image.
BACK SPACE	Go up to the folder above.
< > or arrows	Previous or next page.
F	Show only names that match a pattern.
N	Make a new disk image here.
C	Copy the selected file to another host.
ESC	Back to the main screen.

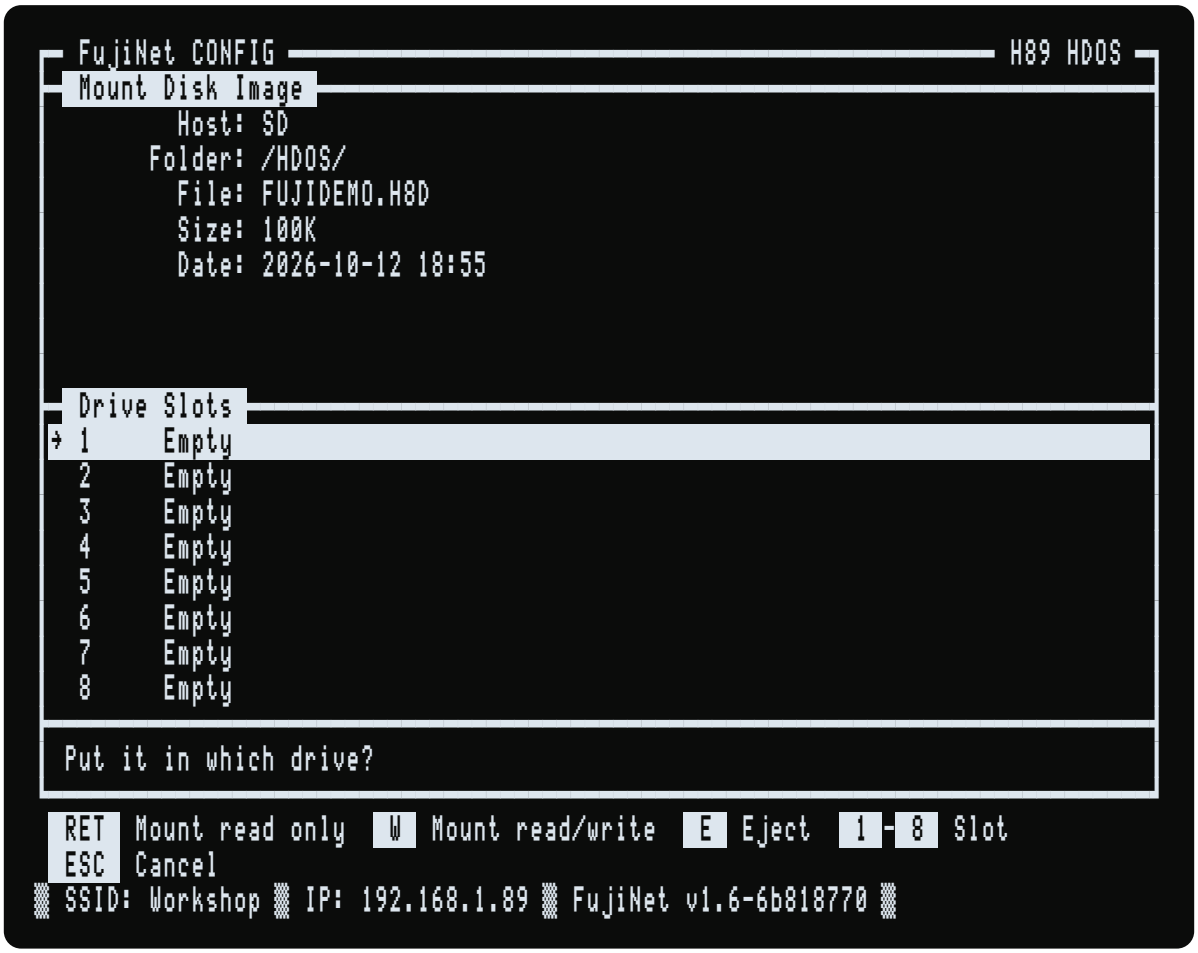
A name marked with a + is a link to another host; RETURN on it takes you there.

SHOWING ONLY SOME NAMES

Push F, type a pattern, and push RETURN. An asterisk stands for any characters: F* shows the names that begin with F, and *.H8D shows the H89 disk images. Folders are always shown, so you can still move about. To show everything again, push F, then **ERASE**, then RETURN.



PICTORIAL 6-3
The microSD card's top folder.



PICTORIAL 6-4
Choosing a drive for a disk image.

PUTTING A DISK IMAGE IN A DRIVE

( ) Browse to the disk image you want, and push RETURN.

CONFIG shows the image's host, folder, name, size and date, and the eight drive slots, and asks `Put it in which drive?`

( ) Pick a drive slot with the arrow keys or its number. Then push RETURN to put the image in read only, or W to put it in read/write.

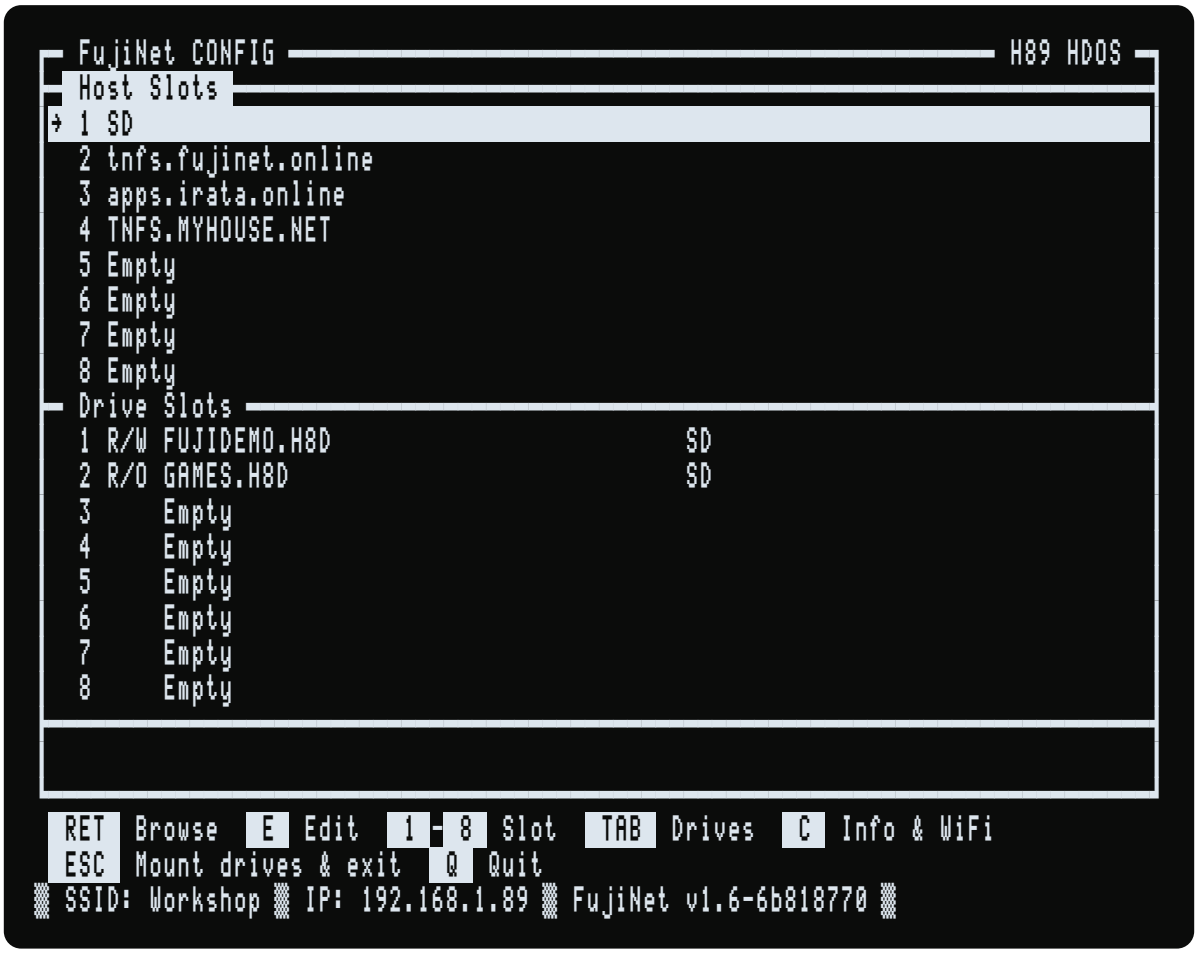
Choose read only unless you mean to write on the disk. A read-only image cannot be harmed, and an image can be read only by more than one drive, or more than one FujiNet, at once. Images on public servers can only be read.

CONFIG takes you back to the folder you were in, so you can choose another image for another drive. The FujiNet notes your choice at once, but the drive is put in service only when you leave CONFIG with **ESC** (see "Leaving CONFIG").

Use .H8D disk images only. The FujiNet drives under HDOS and CP/M read the H-17 diskette format, and refuse anything else (see Page 7-1).

THE DRIVE SLOTS

On the main screen, push **TAB** (or move down past host slot 8) to work in the drive slots. Each line shows the slot's number, R/O or R/W, the disk image's name, and



PICTORIAL 6-5
Two drives in service: one read/write, one read only.

the host it comes from. The information line shows the full name of the selected image.

KEY	DOES
E	Eject: empty the drive slot.
R	Make the drive read only.
W	Make the drive read/write.
ERASE	Eject every drive.
TAB	Back to the host slots.

Under HDOS, dismount a FujiNet drive before you change what is in it (see Page 7-3).

NEW DISK IMAGES

To make a new, empty disk image on a host you can write to (the microSD card, or your own server):

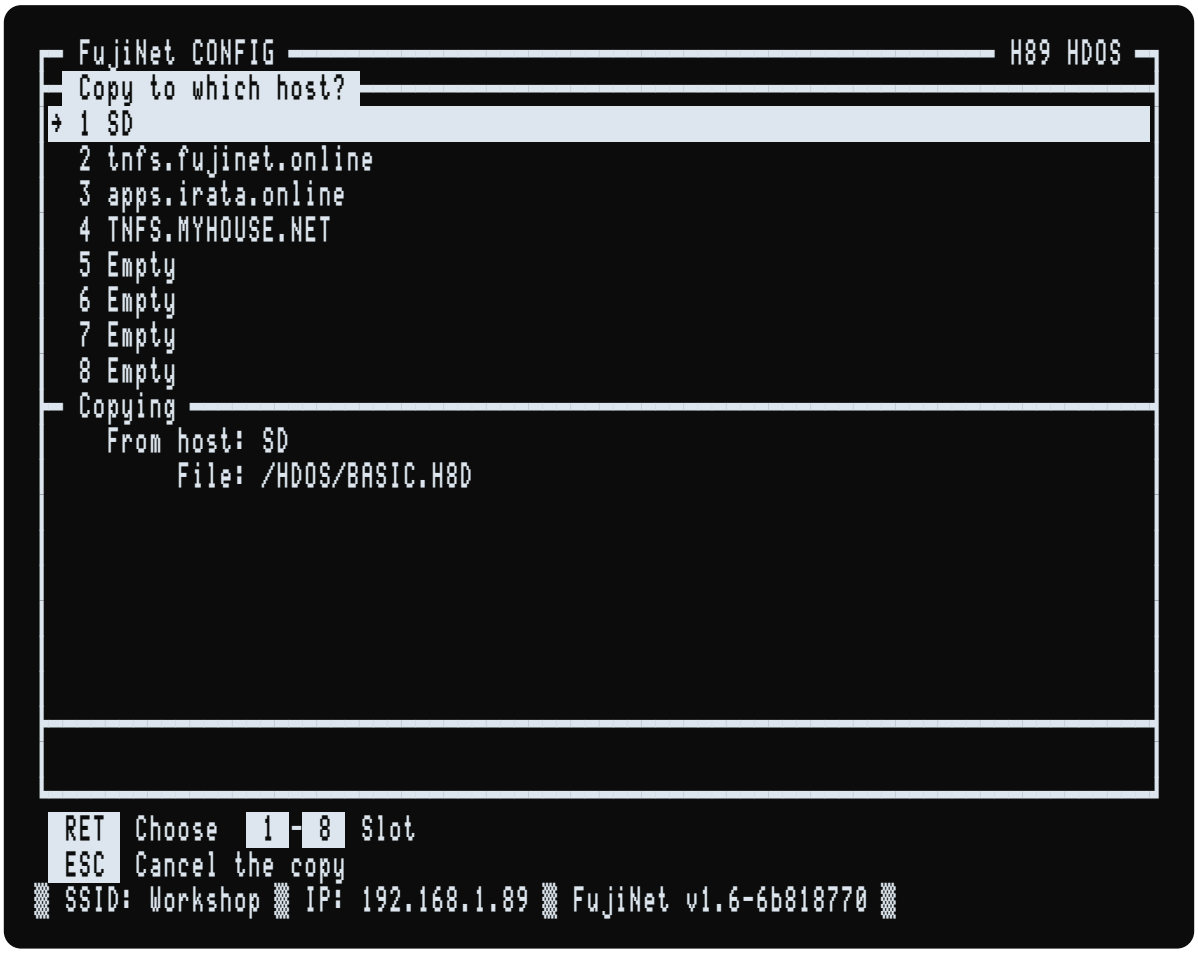
( ) Browse to the folder where it should go, and push N.



( ) Push 1 for a 100K image. That is the size of an H-17-1 diskette and the only size the FujiNet drives use. (The other sizes are for other computers.)

( ) Type a name for the image, ending in .H8D, and push RETURN. Then pick a drive slot for it and push RETURN.

A new image is blank. Before HDOS can use it, it must be initialized; before CP/M can, its directory must be cleared. See “Blank Disk Images” on Page 7-9.



PICTORIAL 6-6
Choosing where a copy goes.

COPYING A FILE

CONFIG can have the FujiNet copy a file from one host to another; the file goes straight from host to host, without passing through the Computer. Use it to bring a disk image from a public server to your microSD card, where you can write on it.

( ) Browse to the file and push C.

( ) CONFIG asks `Copy to which host?` Pick the host to copy to and push RETURN.

( ) Browse to the folder where the copy should go, and push C again.

CONFIG shows `Copying... this can take a while.` and then takes you back to where you started. The host you copy to must be one you can write to.

THE FUJINET'S INFORMATION

On the main screen, push C (Info & WiFi). CONFIG shows the network the FujiNet has joined, its own name and address on the network, the address of your router and name server, its MAC address (which some routers ask for), the version of its firmware, and the versions of CONFIG and the library under it.



The IP address is the one to type into a web browser on another computer to reach the FujiNet's web page (see Page 8-2).

On this screen, S changes the WiFi network (see "Joining Your WiFi Network") and R joins the same network again. Any other key takes you back to the main screen.

LEAVING CONFIG

When the drive slots hold what you want, push **ESC** on the main screen. CONFIG puts each drive in service and marks it OK, or FAIL if the disk image could not be opened. Then it gives the Computer back to HDOS or CP/M and lists the drives:

```
FUJINET DRIVES:                    FujiNet drives:
  1 R/W SD:FUJIDEMO.H8D              1 R/W SD:CPMDEMO.H8D
  2 R/O SD:GAMES.H8D                 A>
>
```

HDOS prints the list in capitals, as it prints everything unless you tell it otherwise. The numbers are the FujiNet's drive slots. Under HDOS, slot 1 is FN0:, slot 2 is FN1:, and so on. Under CP/M, FNDISK shows you which letter each slot has (Page 7-6).

If a drive could not be put in service, CONFIG says `A drive could not be mounted.` and waits for a key. The usual reasons are an image that has gone from its host, a host that cannot be reached, or an image put in read/write on a host that can only be read.

To leave CONFIG without putting drives in service, push Q (Quit) instead. Your choices are kept on the FujiNet for the next time.

WHEN THE FUJINET STARTS

Each time you turn on the Computer, the FujiNet starts afresh. It joins your WiFi network and remembers its host slots and your choices of disk images. But as it comes from the factory, it does not put the drives in service by itself: it waits for CONFIG. So, after you turn on the Computer, run CONFIG and push **ESC** before you use the FujiNet drives.

Pushing **SHIFT** and **RESET** does not restart the FujiNet. Its drives stay in service, and after a reset you can go straight to FN.DVD or FNDISK.

If you would rather the FujiNet put your drives in service by itself, open its web page (Page 8-2), set "Enable CONFIG boot disk" to "No," and save. From then on, after you turn on the Computer, you can go straight to FN.DVD or FNDISK too. CONFIG is still there whenever you want to change something.

USING FUJINET DRIVES

THE FNDISK DRIVERS

Once CONFIG has put disk images in the FujiNet's drive slots, a driver lets the operating system use them. Together, the two drivers are called FNDISK:

DRIVER	FOR	THE DRIVES ARE
FN.DVD	HDOS 2.0	FN0: through FN7:, for drive slots 1 through 8.
FNDISK.COM	CP/M 2.2.04	D: through K: (or eight drives from a letter you choose), for slots 1 through 8.

Each FujiNet drive is a disk image in the H-17-1 format: 40 tracks of 10 sectors of 256 bytes, 102,400 bytes in all, the same as a single-sided H-17 diskette. The drivers check what is in a slot before they use it, and refuse anything that is not an .H8D image, so a disk image for some other computer can do no harm.

A FujiNet drive works like any other drive on your system. You can catalog it, copy files to and from it, and run programs from it. An image on the microSD card answers quickest; one on a server across the Internet answers as quickly as the Internet allows.

The drivers ask the FujiNet for each sector as it is needed, and write each sector back to the disk image at once; nothing is held back waiting to be written.

NOTE: The H89 cannot start HDOS or CP/M from a FujiNet drive. Start your system from its own diskette; the FujiNet drives join it once it is running.

HDOS: FN0: TO FN7:

With FN.DVD on your system diskette (see "Preparing HDOS," Page 5-1), HDOS has eight more drives. Drive slot 1 is FN0:, slot 2 is FN1:, and so on up to slot 8, FN7:. You mount them and dismount them like the drives you already have.

MOUNTING A FUJINET DRIVE

( ) Run CONFIG, put a disk image in drive slot 1, and push **ESC** to leave. (Or, if the FujiNet puts its drives in service by itself, just start HDOS.)

( ) Type MOUNT FN0:↵. HDOS loads FN.DVD if it has not done so already, reads the disk's label and reports it.

( ) Use the drive. For example: CAT FN0:↵, TYPE FN0:HELLO.TXT↵, COPY FN0:PIP.ABS=SY0:PIP.ABS↵.

```
VOLUME 000, MOUNTED ON FN0:
LABEL: FUJINET DEMO DISK
>CAT FN0:
NAME    .EXT    SIZE      DATE         FLAGS   12-OCT-80
HELLO   .TXT    1       11-OCT-80
   1 FILES, USING 1 SECTORS (366 FREE)
>TYPE FN0:HELLO.TXT
HELLO FROM A FUJINET DRIVE ON THE H89
THE SECOND LINE
>COPY FN0:PIP.ABS=SY0:PIP.ABS
1 FILES COPIED
>CAT FN0:
NAME    .EXT    SIZE      DATE         FLAGS   12-OCT-80
HELLO   .TXT    1       11-OCT-80
PIP     .ABS    19      12-OCT-80
   2 FILES, USING 20 SECTORS (346 FREE)
>
```

You may load the driver before you need it with LOAD FN:↵. It stays in memory until you start the system again, and uses about 4.3K.

CHANGING DISKS

HDOS, like any careful operating system, remembers what it knows about a mounted disk. So before you change what is in a FujiNet drive, dismount it, and mount it again afterward:

( ) Type DISMOUNT FN0:↵.

( ) Run CONFIG, change the disk image in slot 1, and push **ESC**.

( ) Type MOUNT FN0:↵.

Do not use HDOS's RESET command on a FujiNet drive. RESET waits for you to change a diskette in a drive, and the FujiNet has no door to open; use DISMOUNT and MOUNT instead.

WHEN HDOS SAYS NO

HDOS reports a FujiNet drive's troubles in its own words:

```
>MOUNT FN1:
?02 UNIT NOT AVAILABLE
>MOUNT FN2:
?02 UNABLE TO READ THIS DISK, IT PROBABLY HAS NOT BEEN PROPERLY INITIALIZED
>MOUNT FN3:
VOLUME 000, MOUNTED ON FN3:
LABEL: FUJINET DEMO DISK
>COPY FN3:PIP.ABS=SY0:PIP.ABS
ERROR ON FILE FN3:PIP.ABS - ?02 DEVICE IS INCAPABLE OF WRITE OPERATIONS (OR IS W
RITE LOCKED)
```

HDOS SAYS	IT MEANS
?02 UNKNOWN DEVICE NAME	HDOS has no FN.DVD: it is not on SY0:, or HDOS has not been started again since you copied it.
?02 UNIT NOT AVAILABLE	That drive slot is empty, the drives have not been put in service since the Computer was turned on (run CONFIG and push **ESC**), or the FujiNet is not answering.
?02 UNABLE TO READ THIS DISK, IT PROBABLY HAS NOT BEEN PROPERLY INITIALIZED	The slot holds something that is not an HDOS disk: another computer's image, a CP/M disk, or a blank image.
?02 DEVICE IS INCAPABLE OF WRITE OPERATIONS (OR IS WRITE LOCKED)	The image is in the slot read only. Dismount, put it in read/write with CONFIG, and mount it again.
?02 READ FAILURE, ?02 WRITE FAILURE	The FujiNet stopped answering, or could not reach the host (for a network host, check the WiFi).

CP/M: FNDISK

Under CP/M, FNDISK.COM adds the FujiNet drives. Run it once each time you start CP/M, after CONFIG has put your disk images in the drive slots. It needs Heath/Zenith CP/M 2.2.04, made into a 62K system (see "Making a 62K CP/M System," Page 5-3).

( ) Start CP/M from your 62K system diskette.

( ) If the drives are not in service yet, run CONFIG (B:CONFIG↵), put your disk images in, and push **ESC**.

( ) Type B:FNDISK↵.

FNDISK installs itself and lists the FujiNet drives, each with its drive slot and the disk image in it:

```
A>B:FNDISK
FNDISK: FujiNet drives for H89 CP/M
Installed.
  D: slot 1  CPMDEMO.H8D (read/write)
  E: slot 2  WORDSTAR.H8D (read only)
  F: slot 3  -
  G: slot 4  -
  H: slot 5  -
  I: slot 6  -
  J: slot 7  -
  K: slot 8  -
A>
```

The drive letters start after the last one your BIOS uses. On an H-17 system, with drives A:, B: and C:, that is D:. A slot with a dash in it is empty. FNDISK stays in memory until you start CP/M again from its diskette (a warm boot, with CTRL-C, keeps it).

You can choose where the letters start by giving FNDISK a letter: B:FNDISK G↵ puts slot 1 on G: and the others after it. All eight must fit by P:, so the latest starting letter is I:, and the letter must not be one of your BIOS's own drives.

USING THE DRIVES

```
A>DIR D:
D: HI      COM : HELLO   TXT
A>TYPE D:HELLO.TXT
Hello from a FujiNet drive on the H89.
The second line.
A>D:HI
Hello from a FujiNet drive on the H89!
A>PIP D:=B:FNDISK.COM
A>DIR D:
D: HI      COM : HELLO   TXT : FNDISK  COM
A>STAT D:
Bytes Remaining On D: 85k
A>
```

Each FujiNet drive holds 92K, with room for 64 directory entries, the same as an H-17 diskette under Heath CP/M.

To see which disk images are in the drives, run FNDISK again. It knows it is already installed, and just lists them:

```
A>B:FNDISK
FNDISK: FujiNet drives for H89 CP/M
Already installed.
  D: slot 1  CPMDEMO.H8D (read/write)
  E: slot 2  WORDSTAR.H8D (read only)
  F: slot 3  -
  G: slot 4  -
  H: slot 5  -
  I: slot 6  -
  J: slot 7  -
  K: slot 8  -
A>
```

CHANGING DISKS

( ) Run CONFIG, change the disk images, and push **ESC**.

CONFIG gives the Computer back to CP/M with a warm boot, and CP/M looks at every drive afresh, so the new disk images are ready to use. There is no need to run FNDISK again unless you have restarted CP/M from its diskette.

WHEN CP/M SAYS NO

CP/M's own error messages tell you when a FujiNet drive cannot do what you ask. An empty drive slot gives

```
A>DIR F:
Bdos Err On F: Select
```

Push RETURN to go on. Writing to a drive whose image is read only gives "Bad Sector":

```
A>PIP E:=B:FNDISK.COM
Bdos Err On E: Bad Sector
```

CP/M waits for a key after a "Bad Sector." Push CTRL-C to give up and go back to the A> prompt; any other key tries the write again, and will fail again.

FNDISK's own messages:

FNDISK SAYS	IT MEANS
Installed.	It is ready; the list follows.
Already installed.	It was installed before; the list follows.
No room above the BIOS. Make a 62K system (MOVCPM17 62 *, then SYSGEN) and boot that.	Your CP/M is a 64K system. See Page 5-3.
Needs Heath/Zenith CP/M 2.2.04.	This is some other CP/M. FNDISK works only with 2.2.04.
No FujiNet card (82C55 at 0D0H) found.	Nothing answered at port 320: the card, its PORT SEL jumper, or another board at 320.
The FujiNet isn't answering.	The card is there but the FujiNet did not reply. Wait a moment and try again; if it still does not answer, turn the Computer off and on. See Page 10-4.
No 8 free drive letters there.	The letter you gave FNDISK is one of your BIOS's drives, or is later than I:.

```
64K HEATH/ZENITH CP/M 2.2.04  09/15/82
FOR H17 DISKS WITH OPTION(S) I
A>B:FNDISK
FNDISK: FujiNet drives for H89 CP/M
No room above the BIOS. Make a 62K system
(MOVCPM17 62 *, then SYSGEN) and boot that.
A>
```

A TYPICAL SESSION

Here is a whole working session, from power-on, under each operating system. These examples assume the FujiNet is waiting for CONFIG after power-on (the way it comes); if you have told it to put its drives in service by itself, skip the CONFIG steps.

HDOS

1. Turn on the Computer and boot HDOS: B↵.
2. MOUNT SY1:↵ (the FujiNet diskette).
3. RUN SY1:CONFIG.ABS↵. Check the drive slots; push **ESC**.
4. MOUNT FN0:↵, and work.
5. Before running CONFIG again: DISMOUNT FN0:↵. (On a 48K Computer, start HDOS again instead.)

CP/M

1. Turn on the Computer and boot CP/M: B↵.
2. B:CONFIG↵. Check the drive slots; push **ESC**.
3. B:FNDISK↵.
4. Use D: through K:.
5. To change disks: B:CONFIG↵, change them, **ESC**.

BLANK DISK IMAGES

The FujiNet drives can only use disk images that have been prepared for their operating system. A new image made with CONFIG's N key is blank, so prepare it first, or make a ready-to-use blank image on your personal computer and copy it to the microSD card.

FOR CP/M

An empty CP/M disk is one whose directory is filled with the byte E5 (hexadecimal). A 102,400-byte file filled with E5 bytes is a ready-to-use empty CP/M disk for FNDISK. On a personal computer with a Unix shell, this command makes one:

```
head -c 102400 /dev/zero | tr '\0' '\345' > BLANK.H8D
```

FOR HDOS

An HDOS disk must be initialized: it needs a label, a directory and its allocation tables. FN.DVD does not initialize disks. Make your blank HDOS images with a disk image tool on your personal computer, or initialize a diskette in your H-17 drive with HDOS's INIT and copy its image. The fujinet-h89 project includes such a tool, `hdosfs.py`:

```
python3 hdosfs.py blank BLANK.H8D --label "MY DISK"
```

IMAGES IN OTHER FORMS

H-17 disk images are also passed around as .H17DISK files, which record the diskette's tracks as the drive saw them. `hdosfs.py convert` turns a single-sided, 40-track .H17DISK into an .H8D for the FujiNet:

```
python3 hdosfs.py convert GAMES.H17DISK GAMES.H8D
```

MORE FROM YOUR FUJINET

THE MICROSD CARD

The microSD card is the FujiNet's own disk: the host called SD. Keep your library of H89 disk images on it, in folders if you like, and put the images you want to write on there, since the public servers can only be read.

- Use a card formatted FAT32. Cards of 32 gigabytes or smaller come that way; larger cards come formatted exFAT, which the FujiNet cannot read, and must be formatted FAT32 first.
- The FujiNet reads the card only when it starts, when the Computer is turned on. Put the card in, or take it out, only with the Computer turned off and no USB cable in the ESP32 module.
- To add disk images, turn off the Computer, take the card out, copy the images onto it with your personal computer, and put it back. Or copy them to it with CONFIG from another host (Page 6-9).
- The FujiNet also keeps a copy of its settings on the card, in a file named fnconfig.ini. Leave it there.

The card is not needed. Without one, the FujiNet works from its network hosts alone, and keeps its settings in its own memory.

THE FUJINET'S WEB PAGE

Once the FujiNet has joined your network, it has a web page of its own. Any computer, tablet or telephone on the same network can show it.

( ) Find the FujiNet's IP address on CONFIG's Info screen (push C on the main screen), or on the 25th line of any CONFIG screen.

( ) On the other computer, open a web browser and type `http://` and the address: for example, `http://192.168.1.89`. Many computers can also find it as `http://fujinet.local`.

The page shows the FujiNet's version and network details, and lets you:

- Give the FujiNet a different name on your network.
- See the host slots and the disk images in the drive slots.
- Set the time zone the FujiNet's clock uses.
- Turn the printer and modem on or off, and choose the printer's type.
- See what has been printed.
- Choose whether the FujiNet puts its drives in service by itself when it starts ("Enable CONFIG boot disk": choose No for that; see Page 6-12).

The WiFi network is changed with CONFIG (Page 6-3), and the firmware with the FujiNet-Flasher (Page 9-1).

Some settings on the page belong to FujiNets for other computers, and do nothing on the H89. The serial port's speed is one: the H89 card talks to the Computer through its 82C55A, not a serial port.

FILE SERVERS

Any host slot can name a server on your own network or on the Internet. Most FujiNet owners use TNFS, the "Trivial Network File System," a small protocol made for computers like the H89. You can run a TNFS server on almost any personal computer: download it from fujinet.online, point it at a folder of disk images, and type the computer's name or address into a host slot. Every FujiNet on your network can then use the images in that folder, and write to them.

The FujiNet can also use folders shared from Windows and other computers by SMB, NFS exports, FTP servers, and folders on web servers. See "Host Slots" on Page 6-5 for how to name each one.

PUBLIC SERVERS

FujiNet owners run servers on the Internet for everyone. Two of them come set up in your host slots:

- tnfs.fujinet.online, the FujiNet project's own server.
- apps.irata.online, a library of programs for many computers.

Each server keeps software for each kind of computer in its own folder; look for the H89 folder. The list of public servers, and whether each is running, is on fujinet.online.

THE FUJINET'S OTHER DEVICES

The FujiNet is more than a set of drives. Like FujiNets for other computers, it carries:

- A printer, which turns what is printed into a file you can see on the web page, laid out as one of several printers of the day.
- A modem, Hayes compatible, which "dials" other computers over the Internet.
- Eight network channels, N1 through N8, that let a program fetch a web page, talk TELNET or SSH, or exchange data with another computer by TCP or UDP.
- A clock, set from the network, for programs that want the date and time.

These are there for H89 programs that use them, and such programs are being written. Watch fujinet.online and the FujiNet Wiki for programs for the H89.

WRITING YOUR OWN

If you write programs, all of the FujiNet is open to you. The Computer talks to the FujiNet by sending small packets through the 82C55A at port 320, and the library that CONFIG and the FujiNet's test programs are built on, fujinet-lib, is free to use from C with the z88dk compiler, for HDOS and for CP/M. The Communication Test in Chapter 3, listed in the Appendix, shows the whole exchange in 114 bytes of 8080 code.

FUJINET GO H89

FujiNet Go H89 is a program for Windows, macOS and Linux that is an H89 with an H19 and a FujiNet built in: the same CONFIG, FNDISK and FN.DVD run on it, and its FujiNet joins the network through your personal computer. It is a pleasant way to try out disk images, or this Manual, away from your H89. It needs copies of the Heath ROMs, which it does not include. The screens in this Manual were made with it.

FIRMWARE UPDATES

ABOUT THE FIRMWARE

The program that runs on the FujiNet itself, on its ESP32, is its firmware. It is improved often, and you can put a new version on your card with the FujiNet-Flasher, a free program for Windows, macOS and Linux, and a USB cable from your personal computer to the ESP32 module on the card. Your settings, host slots and drive slots are kept through an update.

The Flasher does the whole job over the USB cable. The ESP32 module goes into its loading mode when the Flasher asks, and starts the new firmware when the Flasher is done. The Computer takes no part, and has no way to put the FujiNet in loading mode.

YOU WILL NEED

- A personal computer with the FujiNet-Flasher, from fujinet.online/download. On Windows you may also need the driver for the ESP32 module's USB interface (the Silicon Labs CP210x).
- A USB cable that carries data, not just power, from your computer to the ESP32 module's USB connector.

WARNING: Leave the Computer on while you update the firmware: its 82C55A then holds the card's bus transceiver off. Hazardous voltages are present inside a running Computer. Never reach into it. Plug the USB cable into the ESP32 module and lead it out of the cabinet before you turn the Computer on; better still, leave a cable there when you install the card (see Page 3-4).

UPDATING THE FIRMWARE

( ) With the Computer off, connect the USB cable to the ESP32 module and lead it out of the cabinet. Close the cabinet. Leave the other end unplugged for now.

( ) Turn on the Computer and get the H: prompt.

( ) Plug the USB cable into your personal computer and start the FujiNet-Flasher.

( ) In the Flasher, choose the serial port of the ESP32 module, choose the platform "Heathkit H89," and choose the firmware version you want. Then click Flash. The Flasher shows its progress, and says when it is finished. The FujiNet then starts the new firmware.

( ) Unplug the USB cable from your personal computer.

( ) Run CONFIG and push C. The Info screen shows the new firmware's version.

IF THE FLASHER CANNOT CONNECT

The computer cannot see the module, or the module did not go into loading mode. Make sure the cable carries data (try another), and on Windows, that the CP210x driver is installed. Then unplug the cable from your personal computer, turn the Computer off and on, and try once more.

THE FUJINET'S DIAGNOSTIC OUTPUT

While it runs, the FujiNet writes a running account of what it is doing to the ESP32 module's USB connector, at 460,800 baud. If you ask other owners for help with a problem, they may ask you for it. The FujiNet-Flasher can show it (its "Serial Debug Output"), as can any terminal program.

TROUBLESHOOTING

IMPORTANT NOTICE

The FujiNet card, its firmware and its programs were designed to work together, and with Heath's own H89, HDOS 2.0 and CP/M 2.2.04. Other monitors, operating systems and accessories may work, and many owners will gladly help you find out, but the FujiNet project cannot promise it.

There is no factory to send your card to. There is something better: the community of FujiNet owners, many of whom have Heath equipment on their benches. If the charts in this section do not solve your problem, ask on the FujiNet Discord or forums (see the inside front cover). Say which Computer, monitor, operating system and FujiNet firmware you have, what you did, and exactly what the screen said.

GENERAL TROUBLESHOOTING INFORMATION

Most troubles with a new FujiNet are in its installation: a jumper, a card not fully seated, or another accessory answering at the same port. The tests in Chapter 3 find these, so when something is wrong, run the Communication Test (Page 3-5) first. If it passes, the card and the Computer are talking, and the trouble is in the settings or the software.

Remember these things the FujiNet does by design:

- The FujiNet starts when the Computer is turned on, and runs until it is turned off. **SHIFT** and **RESET** reset only the card's 82C55A; the next FujiNet program sets it up again.
- After it starts, the FujiNet waits for CONFIG before it puts its drives in service, unless you have told it otherwise (Page 6-12).
- The FujiNet reads its microSD card only when it starts.
- A USB cable from a personal computer keeps the FujiNet running with the Computer off. Unplug it when you are done with it.
- It hears only 2.4 GHz WiFi networks.
- The H89 cannot start HDOS or CP/M from a FujiNet drive.
- Public servers on the Internet can only be read.

TROUBLESHOOTING CHARTS

The following charts list conditions and their possible causes. Find the condition that matches what you see, and check each cause in turn.

INSTALLATION AND TESTING

CONDITION	POSSIBLE CAUSE
The Computer does not give the H: prompt with the card installed.	The card is not fully seated, or is offset by a pin. Turn off the Computer, unplug it, and reseat the card. Switch SW501 is set for automatic booting or a memory test at power-up: push **SHIFT** and **RESET**.
Port Test reads 377.	Nothing answered at port 322. The PORT SEL jumper is missing or not on 320Q; JP1 is not on 0; the card is not seated; the card is in the floppy disk interface's sockets (P506 and P512).
Port Test reads something other than 200 (or 240), or the second reading is not 252.	Another accessory also answers at port 320 to 327: an HA-88-3 with the 8250 for its 320 port installed. Remove that 8250. Otherwise, a fault on the card (U1, U2).

CONDITION	POSSIBLE CAUSE
Communication Test: only H (and a beep).	As for "Port Test reads 377," above.
Communication Test: 1, and nothing more.	The FujiNet is not taking bytes. Its firmware is missing or is not the H89 firmware for this card; it is still in loading mode after a firmware update (unplug the USB cable from your personal computer, and turn the Computer off and on); no power on the ESP32 module (D2); a fault in U3 or U5 (74LVC245), or in the wires to IO21 and IO22.
Communication Test: 12, and nothing more.	The FujiNet took the request but no answer came back: the wrong firmware, or a fault on the return path (U2's port A strobe lines, U3, U5).
Communication Test: no beep, no H, garbled screen.	The program was keyed in wrong. Check every entry (Page 3-6). Make sure you ran it with G040100.
Communication Test: 12 and the firmware version, but no names.	Normal until the FujiNet has joined a WiFi network. Run CONFIG.

CONFIG AND WIFI

CONDITION	POSSIBLE CAUSE
CONFIG says "Contacting FujiNet…" for a long time, then shows empty slots.	The FujiNet is not answering. Run the Communication Test.
HDOS says "NOT ENOUGH RAM FOR THIS PROGRAM" for CONFIG.ABS.	On a 48K Computer, FN.DVD has been loaded: start HDOS again and run CONFIG first (Page 4-1). Less than 48K of memory. A CONFIG.ABS from some other source: use the one on the FujiNet for HDOS diskette.
No networks are listed, or not yours.	Your network is on 5 GHz only: give the router's 2.4 GHz band its own name. The router is too far away (the antenna is inside the cabinet). The network hides its name: use ⟨Enter a network name⟩.
CONNECT FAILED or UNABLE TO CONNECT.	The password is wrong; capitals count. Your router admits only computers it knows: give it the MAC address from the WiFi Networks screen.
Opening a host flashes "COULD NOT MOUNT HOST SLOT." and goes back.	The host's name is misspelled, the server is not running, or (for SD) there is no microSD card, or it is not FAT32.
"A drive could not be mounted."	The image is gone from its host, the host cannot be reached, or the image was put in read/write on a host that can only be read.
Typing seems to drop letters under CP/M.	Type a little more slowly. CP/M's BIOS keeps no type-ahead while CONFIG redraws the screen.

HDOS DRIVES

CONDITION	POSSIBLE CAUSE
?02 UNKNOWN DEVICE NAME.	FN.DVD is not on SY0:, or HDOS has not been started since it was copied there (Page 5-1).
?02 UNIT NOT AVAILABLE.	The slot is empty; the drives have not been put in service since the Computer was turned on (run CONFIG and push **ESC**); the FujiNet is not answering.
?02 UNABLE TO READ THIS DISK…	The slot holds something that is not an initialized HDOS disk.
?02 DEVICE IS INCAPABLE OF WRITE OPERATIONS…	The image is in read only. Dismount, change it to read/write with CONFIG, mount again.
CAT shows the old disk after you changed it.	The drive was not dismounted before CONFIG. DISMOUNT and MOUNT it.

CP/M DRIVES

CONDITION	POSSIBLE CAUSE
No room above the BIOS.	Make a 62K system (Page 5-3).
Needs Heath/Zenith CP/M 2.2.04.	FNDISK works with CP/M 2.2.04 only.
No FujiNet card (82C55 at 0D0H) found.	See "Port Test reads 377," above.
The FujiNet isn't answering.	The FujiNet is still starting: wait a moment and run FNDISK again. Or it is in loading mode after a firmware update: unplug the USB cable from your personal computer, and turn the Computer off and on.
Bdos Err On D: Select.	The slot is empty, holds something that is not an .H8D image, or the drives have not been put in service since the Computer was turned on (run CONFIG).
Bdos Err On D: Bad Sector, when writing.	The image is in read only. Push CTRL-C; change it to read/write with CONFIG.
Drives D: through K: are gone.	CP/M was started from its diskette (or the Computer reset) since FNDISK ran. Run FNDISK again.

FIRMWARE UPDATES

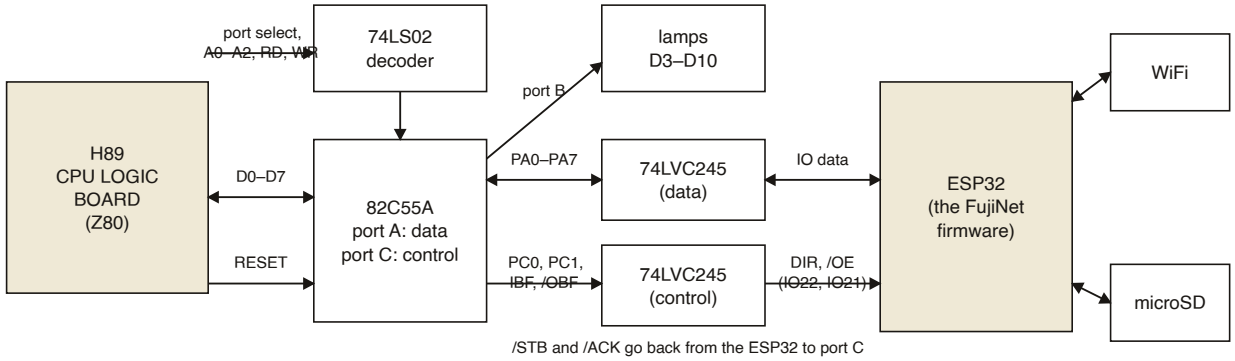
CONDITION	POSSIBLE CAUSE
The Flasher cannot connect.	The USB cable carries only power; the CP210x driver is missing (Windows); the Computer is off.
After an update, the programs say the FujiNet is not answering.	It is still in loading mode. Unplug the USB cable from your personal computer, and turn the Computer off and on.

CIRCUIT DESCRIPTION

GENERAL

Refer to Pictorial 11-1 while you read this description.

The card joins two different worlds. On one side is the H89's accessory bus: five-volt logic, an 8-bit data bus, and a set of decoded port-select lines from the CPU logic circuit board. On the other is the ESP32, a 3.3-volt microcontroller that runs the FujiNet firmware and has the WiFi radio built in. An 82C55A programmable peripheral interface stands between them. The Computer reads and writes it as four I/O ports; the ESP32 works its handshake lines directly.



PICTORIAL 11-1
How the card connects the H89 to the FujiNet.

ADDRESS DECODING

The CPU logic circuit board decodes the I/O address and gives each group of eight ports a select line on the accessory bus. The PORT SEL jumper (J1) connects one of four of these, for the groups at 320, 330, 340 and 170, to the card. The 74LS02 (U1) combines the chosen select line with address line A2, which JP1 passes straight through (on 0) or inverted (on 4), so the 82C55A's chip select is active for the first four ports of the group or the last four. Address lines A0 and A1 choose among the 82C55A's port A, port B, port C and control register. The Computer's RD and WR strobes go straight to the 82C55A.

U1 also inverts the bus's RESET line for the 82C55A, which resets whenever the Computer does.

THE 82C55A

The 82C55A is U2, a PLCC part, or U6, a 40-pin DIP in the same place; only one is fitted. The programs put it in Mode 2 by writing 300 octal to its control register. In Mode 2, port A is a bidirectional, strobed port: a byte the Computer writes is held in port A's output latch until the other side takes it, and a byte from the other side is held in the input latch until the Computer reads it. Port C carries the handshake:

BIT	NAME	USE
PC7	/OBF	Low while a byte the Computer wrote is waiting for the ESP32 (IO35).
PC6	/ACK	From the ESP32 (IO15): it has taken the byte.
PC5	IBF	High while a byte from the ESP32 (IO34) is waiting for the Computer.
PC4	/STB	From the ESP32 (IO4): here is a byte.
PC3	INTR	Interrupt request, through Q1 to J3. Not used (J3 open).
PC2	–	Not connected (see "The Rework").
PC1	/OE	Output: turns on U3, the data transceiver. The ESP32 reads it on IO21.
PC0	DIR	Output: U3's direction, 1 toward the ESP32. The ESP32 reads it on IO22.

To send a byte, a program sets DIR to 1, writes the byte to port A, and waits for /OBF to go high again, which shows the ESP32 took it. To receive, it sets DIR to 0 and waits for IBF, then reads port A. The Communication Test (in the Appendix) is a complete example.

Port B drives the eight lamps, D3 through D10, through the resistor network RN1: a 0 bit lights its lamp. It is also brought out on J12. The FujiNet does not use it.

LEVEL SHIFTING

Port A's eight data lines reach the ESP32 through U3, a 74LVC245 bus transceiver powered from 3.3 volts. Its inputs tolerate five volts, so it passes the 82C55A's five-volt outputs safely to the ESP32, and the ESP32's 3.3-volt outputs are high enough for the 82C55A's inputs. PC0 sets its direction and PC1 turns it on. R1 holds PC1 high, and so U3 off, while the 82C55A's ports are inputs after a reset.

U5, a second 74LVC245, wired to pass one way only, brings DIR, /OE, IBF and /OBF down to 3.3 volts for the ESP32. /STB and /ACK go from the ESP32 to the 82C55A directly. J2 brings the handshake lines out for a logic analyzer.

THE REWORK

The card is a V1.1 board reworked for the present firmware. The ESP32's EN, IO0 and IO2 are no longer connected to the card, and LED D1 and its resistor R7, on IO22, are removed. Two wires take /OE from U5 to IO21, and DIR to IO22. So nothing the Computer does can start, stop or restart the ESP32, or put it in its loading mode, and IO0 and IO2, which the ESP32 reads as it starts, are left to the module.

WHEN THE FUJINET STARTS

The ESP32 starts when the Computer is turned on, or when a USB cable brings it power, and runs the firmware in its flash. When the Computer resets, the 82C55A's ports all become inputs and port A leaves Mode 2, but the ESP32 is not touched.

A program sets the card up with one mode word, 300, which puts port A in Mode 2 and makes port B and PC0 through PC2 outputs, all low: U3 on and facing the Computer, and every lamp lit. Then it reads port A once, to drop any byte left from before. A mode word touches only the 82C55A, so a program may send one whenever it starts.

THE ESP32 AND THE MICROSD CARD

The ESP32 module, U4, is an ESP32-DevKitC-VE. It carries the processor, its flash memory and PSRAM, the WiFi radio and antenna, and a USB-to-serial interface for firmware loading and diagnostic output, with the EN and BOOT buttons that go with it. The microSD breakout board plugs into J6. Its socket is wired to the ESP32's SPI port (IO5, IO18, IO19 and IO23), with 10K pull-up resistors (R3 through R6).

POWER

The card takes its power from the accessory bus. The 82C55A and the 74LS02 run from the bus's five volts, which also reach the ESP32 module through the diode D2. The module's own regulator makes 3.3 volts for the ESP32, U3, U5 and the microSD card. D2 keeps a USB cable's power off the bus.

PARTS LIST

The circuit component numbers are those printed on the card (V1.1). Parts marked with an asterisk (*) should be confirmed against your card's own markings. The rework (Chapter 11) removes D1 and R7 and adds two wires.

INTEGRATED CIRCUITS

COMP. NO.	DESCRIPTION
U1	74LS02 quad 2-input NOR gate, 14 pins
U2*	82C55A programmable peripheral interface, 44-pin PLCC; or U6
U6*	82C55A, 40-pin DIP, in U2's place
U3	74LVC245 octal bus transceiver, 20 pins (data; under the ESP32 module)
U5	74LVC245 octal bus transceiver, 20 pins (control lines)

MODULES

COMP. NO.	DESCRIPTION
U4	ESP32-DevKitC-VE module (ESP32-WROVER-E, 8 MB flash)
–	microSD card breakout board, on J6

RESISTORS

COMP. NO.	DESCRIPTION
R1	10K (U3's /OE held high)
R2	10K (Q1's base)
R3–R6	10K (microSD pull-ups)
R7	560 Ω (LED D1). Removed in the rework.
RN1	1.2K network, 8 resistors (the lamps)

CAPACITORS

COMP. NO.	DESCRIPTION
C1	47 μF electrolytic
C2–C4	0.1 μF

SEMICONDUCTORS AND INDICATORS

COMP. NO.	DESCRIPTION
D1	LED, red ("FujiNet_IO"). Removed in the rework.
D2	1N5819 Schottky diode
D3–D10	LEDs, the lamps for port B (PB0–PB7)
Q1	2N3904 transistor

CONNECTORS AND JUMPERS

COMP. NO.	DESCRIPTION
J1	PORT SEL: 2 × 4 pin header, with one jumper
JP1	BA2: 3-pin header, with one jumper
J2	Logic analyzer header, 10 pins
J3	/INTE_ENA: 2 × 3 pin header
J4, J7, J8*	Rows of pads for the 10-pin bus connector (data D0–D7)
J5, J9, J11*	Rows of pads for the 25-pin bus connector (power, address, control, port selects)
J6	microSD breakout header, 6 pins
J12	Port B header, 10 pins
–	Rework wires: /OE to IO21, DIR to IO22

APPENDIX

QUICK REFERENCE

CONFIG: THE MAIN SCREEN

KEY	DOES
arrows, **1–8**	Pick a host slot or drive slot
RETURN	Browse the host (edit an empty slot)
E	Edit a host slot; eject a drive slot
R / W	Drive slot: read only / read/write
ERASE	Eject all drives
TAB	Host slots / drive slots
C	Info and WiFi
ESC	Put the drives in service and leave
Q	Leave without putting them in service

CONFIG: BROWSING

KEY	DOES
RETURN	Open a folder; choose an image
BACK SPACE	Up a folder
< >	Previous / next page
F	Filter (* for any characters)
N	New disk image
C	Copy to another host
ESC	Back to the main screen

In "Put it in which drive?": **RETURN** read only, W read/write.

HDOS

TYPE	TO
`RUN SY1:CONFIG.ABS`	Run CONFIG
`LOAD FN:`	Load FN.DVD now (optional)
`MOUNT FN0:`	Use drive slot 1
`DISMOUNT FN0:`	Before CONFIG changes slot 1
`CAT FN0:`	List its files

CP/M

TYPE	TO
`B:CONFIG`	Run CONFIG
`B:FNDISK`	Add drives D: to K:, or list them
`B:FNDISK G`	Add them from G:
`MOVCPM17 62 *`	Then SYSGEN: make a 62K system

THE MONITOR

TYPE	TO
`S0040100`↵	Substitute: SPACE next, - back, RETURN done
`G0040100`↵	Run the program at 040100
`O323,300`↵	(MTR-90) Set up the 82C55A
`I322`↵	(MTR-90) Read port C: 200
`O321,252`↵	(MTR-90) Light every other lamp
SHIFT RESET	Reset; the FujiNet keeps running

THE COMMUNICATION TEST

This is the program you key in for the Communication Test (Page 3-5), as the assembler listed it. The first column is the address, in split octal; the second, the numbers you enter. The program uses the monitor's routine at 003302 to print a character, and ends with a halt, which the monitor turns into a beep and its prompt.

```
                    ;       ACCTEST - H89 FUJINET ACCEPTANCE TEST
                    ;
                    ;       KEYED INTO THE MONITOR AT 040.100 AND RUN WITH G040100.
                    ;       SETS UP THE CARD, ASKS IT FOR ITS ADAPTER CONFIGURATION AND
                    ;       PRINTS THE ANSWER.  PRINTS 1 WHEN THE 82C55 ANSWERS, 2 WHEN
                    ;       THE ESP32 HAS TAKEN THE REQUEST, THEN THE REPLY'S TEXT.  ENDS
                    ;       WITH A HALT, WHICH THE MONITOR TURNS INTO A BEEP, AN H AND
                    ;       THE H: PROMPT.
                    ;
                    ;       NUMBERS ARE OCTAL (Q) OR SPLIT OCTAL (A), AS IN HEATH LISTINGS.
                    ;
                    ;       Copyright (C) 2026 Thomas Cherryhomes
                    ;       SPDX-License-Identifier: GPL-3.0-or-later
                    ;
                    PDATA   EQU     320Q            ;82C55 PORT A, THE BYTE PIPE
                    PSTAT   EQU     322Q            ;82C55 PORT C: STATUS IN, CONTROL OUT
                    PCTRL   EQU     323Q            ;82C55 CONTROL WORD
                    WCC     EQU     003302A         ;MONITOR: WRITE CONSOLE CHARACTER
                    ;
                            ORG     040100A
                    ;
040100  076 300     START:  MVI     A,300Q          ;PORT A MODE 2, PC0-PC2 OUT, ALL 0
040102  323 323             OUT     PCTRL           ; (THE FUJINET IS ALREADY RUNNING)
040104  333 322             IN      PSTAT           ;IS THE 82C55 THERE?
040106  346 007             ANI     007Q            ; PC2-PC0 MUST READ BACK 000
040110  302 251 040         JNZ     DONE            ;NO: HALT NOW
040113  076 001             MVI     A,001Q          ;245 ON (/OE=0), DIR TOWARD ESP32
040115  323 322             OUT     PSTAT
040117  076 061             MVI     A,'1'           ;CHECKPOINT 1
040121  315 302 003         CALL    WCC
040124  041 252 040         LXI     H,REQ           ;SEND THE REQUEST
040127  006 010             MVI     B,8
040131  176         SEND:   MOV     A,M
040132  323 320             OUT     PDATA
040134  333 322     WOBF:   IN      PSTAT           ;WAIT UNTIL THE ESP32 HAS IT
040136  346 200             ANI     200Q            ; (/OBF HIGH AGAIN)
040140  312 134 040         JZ      WOBF
040143  043                 INX     H
040144  005                 DCR     B
040145  302 131 040         JNZ     SEND
040150  076 000             MVI     A,000Q          ;DIR TOWARD THE H89
040152  323 322             OUT     PSTAT
040154  076 062             MVI     A,'2'           ;CHECKPOINT 2
040156  315 302 003         CALL    WCC
040161  001 000 005 SYNC:   LXI     B,002400Q       ;B=5 HEADER BYTES TO SKIP, C=0
040164  315 237 040 HEAD:   CALL    GETB            ;FIND THE FRAME, SKIP THE HEADER
040167  376 300             CPI     300Q            ; (START OVER ON EVERY C0)
040171  312 161 040         JZ      SYNC
040174  005                 DCR     B
040175  302 164 040         JNZ     HEAD
040200  315 237 040 SHOW:   CALL    GETB            ;PRINT THE REPLY UP TO THE END C0
040203  376 300             CPI     300Q
040205  312 251 040         JZ      DONE
040210  376 041             CPI     041Q            ;PRINTABLE?
040212  332 222 040         JC      GAP
040215  376 177             CPI     177Q
040217  332 230 040         JC      PUT
040222  076 040     GAP:    MVI     A,040Q          ;ANYTHING ELSE IS A SPACE,
040224  271                 CMP     C               ; BUT ONLY ONE IN A ROW
040225  312 200 040         JZ      SHOW
040230  117         PUT:    MOV     C,A
040231  315 302 003         CALL    WCC
040234  303 200 040         JMP     SHOW
                    ;
040237  333 322     GETB:   IN      PSTAT           ;WAIT FOR A BYTE FROM THE ESP32
040241  346 040             ANI     040Q            ; (IBF HIGH)
040243  312 237 040         JZ      GETB
040246  333 320             IN      PDATA
040250  311                 RET
                    ;
                    ;       THE HALT COMES LAST: THE MONITOR'S CLOCK SEES A HALT WHEN THE
                    ;       BYTE BEFORE THE INTERRUPTED ADDRESS IS 166Q, SO NO CODE MAY
                    ;       FOLLOW ONE.
040251  166        DONE:    HLT                     ;BACK TO THE MONITOR
                    ;
                    ;       GET ADAPTER CONFIG (350Q) FOR THE FUJINET DEVICE (160Q),
                    ;       A SIX-BYTE FUJIBUS HEADER BETWEEN SLIP END BYTES (300Q).
040252  300 160 350 REQ:    DB      300Q,160Q,350Q,006Q,000Q,137Q,000Q,300Q
        006 000 137
        000 300
                    ;
                            END     START
```

THE FUJINET'S DEVICES

Inside the FujiNet are a number of devices, each with an identifying number. A program reaches one by sending a packet through port 320: the device's number, a command, the packet's length, a check byte, a byte describing what follows, and then any parameters and data. Each packet travels between two bytes of 300 octal (C0 hexadecimal), and the FujiNet answers with a packet of its own.

DEVICE	IS
70 hex	The FujiNet itself: WiFi, hosts, slots, copying
31–38 hex	Disk drives, slots 1 through 8
71–78 hex	Network channels N1 through N8
40 hex	Printer
45 hex	Clock
50 hex	Modem

The Communication Test sends the FujiNet device (70 hex) command E8 hex, "get adapter configuration," whose answer holds the WiFi network's name, the FujiNet's name and addresses, and its firmware version. The full description of every device and command is in the FujiNet Wiki and in the FujiNet Network Protocol Handbook.

GLOSSARY

Disk image – A file holding everything on a diskette, sector by sector. The FujiNet's drives use .H8D images.
Drive slot – One of the FujiNet's eight drives. Under HDOS, FN0: to FN7:; under CP/M, D: to K:.
Firmware – The program that runs on the FujiNet's ESP32.
FNDISK – The FujiNet's drivers: FN.DVD for HDOS, FNDISK.COM for CP/M.
H8D – The disk image format for the H-17-1 diskette: 102,400 bytes.
Host – A place disk images come from: the microSD card, or a server.
Host slot – One of the FujiNet's eight host names.
MAC address – The FujiNet's own hardware address on the network.
Mode 2 – The 82C55A's bidirectional, strobed mode, used by the card.
Mount – Under HDOS, make a disk known to the system. In CONFIG, put a disk image in service.
Read only, read/write – Whether a drive slot may be written to.
SSID – A WiFi network's name.
Split octal – The Heath way of writing a 16-bit address as two octal bytes: 040100.
TNFS – Trivial Network File System, the file-server protocol most FujiNet hosts use.

GETTING HELP

- fujinet.online: news, downloads, and the list of public servers.
- The FujiNet Discord (discord.gg/7MfFTvD): other owners, at any hour.
- github.com/FujiNetWIFI: the source of the firmware, CONFIG and the drivers, where you can report a problem.
- github.com/FujiNetWIFI/fujinet-manuals: report a mistake in this Manual.

When you ask, say which Computer, monitor (MTR-88, –89 or –90), operating system and FujiNet firmware you have, what you did, and exactly what the screen said.

MY FUJINET

WiFi network (SSID) _____

FujiNet's IP address _____

FujiNet's name _____

MAC address _____

Firmware version _____

Card installed in sockets _____

Date installed _____

HOST SLOTS

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

FAVORITE DISK IMAGES

_____

_____

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_____

FUJINET

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